Nodal-Precession Phasing for Multi-Object Debris Removal in Low Earth Orbit

An Audited and Reproducible Re-evaluation of Waiting below the Targets, with Exact Optimiser Checks, a Servicer Design and Cost Assessment, a Comparison with Removal Demonstrations, and a Full Engineering Analysis of the Original Nanotug Concept

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Abstract

Massive derelict rocket bodies in a few low Earth orbit bands dominate the future fragment supply, and a single servicing spacecraft could visit many of them if it lets the J_2 nodal precession rotate the orbit planes into line instead of paying for plane changes. This paper re-derives, tests and re-runs an earlier draft study of that idea on one frozen public snapshot (Celestrak catalogue and element sets of 18 August 2026, common epoch 15:02:52 UTC, hashes published) and on the recovered 2025 inputs, and it adds the subsystem design, cost and policy view that a real servicer needs. The draft's catalogue counts are reproduced exactly from the 2025 files: the 1,288 "actionable" objects are the non-operational payloads and rocket bodies at 600–1,500 km mean altitude in a March 2025 catalogue, 193 of which had already decayed, and the 593 objects in 40 sub-clusters follow once April 2025 node angles are joined; on the 2026 snapshot the same rule restricted to objects in orbit gives 1,151. The "lower is better" note rests on a Hohmann cost that is not symmetric, and the planner's scalarisation weight was never tuned. We prove that the drift-rate gain of lowering exceeds that of raising for every offset ($R(x) > 1$, series $1 + \frac{9}{2}x + \frac{81}{8}x^2 + \dots$), that the cost ratio is $1 + \frac{3}{2}x$ and that the net gain per ΔV is $1 + 3x$, and we give a decision rule, an eccentricity reversal and error bounds. Against an exact dynamic-programming optimum on 32 small instances, a tuned beam search is 1.9% (95% CI 0.6–3.4) from optimal, while the draft's configuration is 10.3% from it. On the 159-object SL-8 pool the median tour reaches 29 objects at 2 km/s and 7 yr, and 67 at 5 km/s and 15 yr, compared with 9 and 19 for impulsive plane changes and 6 and 12 for waiting without an altitude change. Cowell and SGP4 checks close the node to $\leq 0.05°$. Of ten novel ideas screened, only a parked carrier with expendable kits is supported (4.5–6 times faster kit placement than a serial tour). Every flown or planned removal demonstration handles one object; a chemical carrier parked 400 km below the SL-8 cluster releases 16 kits within about 310 days of launch, but kits that carry their own de-orbit propellant raise its launch mass from 3.3 t to 4.8–5.9 t. A parametric estimate after the SMAD cost models puts this 16-kit mission at 566 M USD (2024 dollars; 10–90%: 396–736 M), or about 35 M USD per object and 25,000 USD per kilogram removed, with most of the cost in development rather than in operations or launch. Electric propulsion saves propellant only on the long visit-only tour, where it needs 1.1 to 3.1 demonstrated thruster lifetimes. The main drawback of the approach is that its advantage is bought with years of mission time, and we claim no new astrodynamics. We then analyse the original concept itself, an ESPA-class electric carrier with 8U nanotugs, using a moving ephemeris, real thrust and eclipse limits, a dispenser design, tug budgets, adhesive contact and rigid-body detumble models and a mission Monte Carlo. The tour closes (19 objects, 2,000 m/s including insertion and disposal, 6.6 yr) and 40 tugs fit in tandem tubes, not in commercial canisters (20). The removal does not close: a pair of tugs gives a 1.4 t stage 54–74 m/s, which leaves a life of hundreds of years, and a 25-yr life needs 5–7

tugs per stage. The concept closes as stabilise and dodge, which is what the nano-tug paper proposed, for about 11.9 objects per mission with spares, or as removal with paired electric kits that are not 8U. The bonded surface of a decades-old stage is the largest unknown.

Nomenclature

Orbit and Perturbation Parameters	
a, e, i	semi-major axis [km], eccentricity, inclination [deg]
p	semi-latus rectum, $a(1-e^2)$ [km]
n	mean motion [rad/s]
μ, R_E, J_2	Earth gravitational parameter, equatorial radius, second zonal harmonic
Ω, $\dot{\Omega}$	right ascension of the ascending node (RAAN) [deg], secular nodal rate [deg/day]
K	nodal-rate constant, $\frac{3}{2}J_2R_E^2\sqrt{\mu}$
Phasing Parameters	
x	relative altitude offset, $\Delta a/a_T$
$R(x)$	ratio of nodal-rate gain for lowering to that for raising
g, g^*	nodal gap [deg], gap threshold of the decision rule [deg]
h	drift-orbit depth below (or above) the target [km]
Δt	dwell time to close a nodal gap [d]
Mission Parameters	
ΔV	velocity change [m/s]
v, v_c	orbital speed, circular speed [m/s]
I_{sp}	specific impulse [s]
C_dA/m	ballistic parameter [m^2/kg]
$w_{\Delta V}$, w_t	planner scalarisation weights on ΔV and time
K (kits)	number of expendable kits placed by the carrier
Servicer Parameters	
P_e, P_d, P_{sa}	load in eclipse and in daylight, solar-array output required [W]
T_e, T_d	eclipse and daylight duration per orbit [min]
X_e, X_d	power-path efficiencies in eclipse and in daylight
I_d, L_d	inherent degradation and life degradation of the solar array
C_r, DOD	battery capacity [W h], depth of discharge
λ	series failure rate of the servicer [1/yr]
h_p	perigee altitude after a de-orbit burn [km]
L, S	learning-curve factor, learning-curve slope [%]
σ	standard deviation of a cost estimate [%]
Nanotug Parameters	
R	expected fragments $\geq$10 cm over 25 yr
λ (Sec. 11)	catastrophic collision rate of an object [1/yr]
ρ	distance of the pad from the spin axis [m]
H, I_t, I_a	angular momentum [N m s], transverse and axial inertia [kg m^2]

1 Introduction

Collision-risk studies since Kessler and Cour-Palais [39] and the long-term projections of the space agencies [45, 47–49] agree that massive intact derelicts, above all rocket bodies concentrated in a few inclination–altitude bands, dominate the future fragment supply [1, 53]. Active removal is

expensive per object, so one launch should remove many. The obstacle is the orbit plane. Objects in one band share inclination but not right ascension of the ascending node (RAAN), and a direct plane change of $\Delta\Omega = 1^\circ$ at $i = 83^\circ$ costs 127.5 m/s at the SL-8 orbital speed; Braun et al. [5] identify plane change as the main obstacle to multi-target removal. The remedy, already used in constellation deployment [16, 40, 52] and in debris tours [9, 31, 66], is to let the secular J_2 drift of the node do the work: the spacecraft waits on an orbit whose nodal rate differs from that of the target.

This paper started from a draft study that proposed waiting *below* the targets (lowering) and reported tours of 24–53 objects. However, the draft's numbers were never independently reproduced, its analytic claims were not checked, and its planner was never compared with an exact optimum. The goal of this paper is to address this gap. This study turns the draft into a reproducible evaluation to a publishable standard, and tests by calculation whether further ideas help. The contributions are evidential, not conceptual (Section 13.2):

1. an independent engine that reproduces the draft's tour exactly, and an audit of every analytic claim and pipeline stage (Section 3);
2. a frozen, hash-pinned snapshot that resolves the catalogue-count discrepancies;
3. exact and sharp statements on lowering versus raising (Section 4);
4. an exact-optimum validation of the sequencing heuristics and a correction of the planner's weight (Section 6);
5. fair baselines, high-fidelity verification, Monte Carlo uncertainty, a caveated collision-risk benefit and a screen of ten novel ideas;
6. an exact reproduction of the draft's 2025 catalogue pipeline from the original inputs, which explains every count it reported (Section 3.1);
7. a servicer design at the level SMAD expects for a mission concept [43], placed against the flown and planned removal demonstrations, with propulsion, power, mass, communications, reliability, de-orbit kit, timeline and cost estimates, and the regulatory context (Sections 10 and 12).
8. a full engineering analysis of the original nanotug concept, covering risk-based targets, a moving ephemeris with in-plane phasing, the low-thrust chaser, the real dispenser, tug budgets, attachment, detumble and mission closure, with a plain statement of what closes (Section 11).

In-orbit servicing and removal have moved from studies to flight in the last decade. RemoveDEBRIS tested a net and a harpoon on its own CubeSat targets in 2018 [4, 20], ELSA-d released and recaptured its own magnetically prepared client in 2021 [4], ADRAS-J approached a non-cooperative upper stage in 2024 [34, 35], and GEO life-extension servicers have docked to operational clients since 2020 [4]. ClearSpace-1 [22, 23] and ADRAS-J2 [34] are to capture one object each later this decade. All of these handle one object per mission. The question of what a multi-object tour would look like as a spacecraft, and what it would cost, is therefore the natural next step for this paper.

The rest of the paper is structured as follows. Section 2 develops the approach: the secular nodal-rate and cost models, the frozen snapshot, the leg and tour model, the solvers, the baselines and the uncertainty and risk models. Section 3 reports the audit of the draft, Section 4 the exact results on lowering versus raising, and Sections 5 and 6 the validation of the numerical model and of the optimiser. The results are then presented for the fair baselines, the headline missions and the collision-risk benefit (Section 7), followed by uncertainty and sensitivity (Section 8) and the

screen of novel methods (Section 9). Section 10 turns the tour into a servicer design and compares it with the demonstrations, and Section 12 gives the regulatory context. The discussion covers the implications, the novelty assessment, the sources of error and model limitations, and the objections a reviewer would raise (Section 13). The paper concludes with the main findings and the work that remains.

2 Approach

The goal of this study is to quantify how many objects a single chaser can visit in one band when the plane changes are done by J_2 nodal drift, and how that compares with the alternatives at equal budget. The analysis links

1. a secular nodal-rate model and exact transfer costs (Section 2.1),
2. a frozen catalogue snapshot and a leg model that turns each target visit into a transfer, a dwell and a final transfer, and
3. a tour optimiser checked against an exact dynamic program,

to compute the number of objects reached, the ΔV, the mission time and the propellant mass, together with their uncertainty.

2.1 Secular nodal rate and transfer costs

Secular nodal rate. For semi-major axis a, eccentricity e, inclination i and $p = a(1-e^2)$, the J_2 secular rate is [6, 41, 42]

$$\dot{\Omega} = -\tfrac{3}{2} J_2 n \Big(\frac{R_E}{p}\Big)^2 \cos i, \qquad n = \sqrt{\mu/a^3}, \tag{1}$$

where n is the mean motion, μ the Earth's gravitational parameter and R_E its equatorial radius. For $e = 0$ the rate reduces to $\dot{\Omega} = -K \cos i \; a^{-7/2}$, with $K = \frac{3}{2} J_2 R_E^2 \sqrt{\mu}$. A target whose rate is higher than the chaser's, in the direction of the gap, is reached after $\Delta t = g/|\dot{\Omega}_c - \dot{\Omega}_t|$ for a gap g. For absolute rates we use the second-order closed form (terms in J_2^2 and J_4) with the SGP4 constants, which reproduces SGP4's own mean nodal rate to 0.08%; Eq. (1) alone is 0.29% high on the SL-8 pool. However, only rate *differences* matter for phasing, so this common-mode bias cancels.

Costs. Hohmann transfers use the exact two-burn expression. A node error of θ_Ω at inclination i costs $2v \sin(\frac{1}{2}\vartheta)$, with $\sin\frac{1}{2}\vartheta = \sin i \, \sin\frac{1}{2}\theta_\Omega$ (equal inclinations). Drag make-up is computed with a tabulated exponential atmosphere and a ballistic parameter $C_d A/m = 0.01\,\mathrm{m}^2/\mathrm{kg}$ (stated, unverified). Propellant follows the rocket equation for two stated designs, both of which are assumptions: a storable bipropellant design ($I_{sp} = 320\,\mathrm{s}$, 700 kg dry) and a Hall-thruster design (1800 s, 900 kg dry).

2.2 Frozen snapshot and pools

The analysis uses *snapshot v3-2026-08-18*, which consists of the Celestrak SATCAT csv (70,292 rows) and the GP element sets for five pools, fetched by the original study on 18 August 2026 from public, credential-free sources; the SHA-256 hashes are in `data/SNAPSHOT.json`. The snapshot was not re-downloaded, because the draft's tours are defined on it and a new epoch would break the comparison. All elements are propagated with SGP4 to one common epoch (2026-08-18T15:02:52.746Z for the SL-8 set; TLE age median 0.14 d, maximum 0.87 d). The pools are SL-8

rocket bodies ($i \approx 82.9°$, 950–1,010 km, 159 objects), sun-synchronous objects (135), Iridium-class objects (28) and SL-16 rocket bodies (20). Nodal rates are evaluated on *mean* elements. The draft used osculating elements, which differ in a by 6.4 km (1σ), a difference comparable to the pool's own spread of 7.5 km. The original 2025 inputs of the draft (two public catalogue files of March and April 2025, element sets of April 2025 and the outputs of its MATLAB and Python stages) were recovered from the author's records and copied read-only into `data/orig2025/`; they are used only to reproduce the draft's counts (Section 3.1).

2.3 Leg model and tour problem

A leg takes the chaser from object c (in its plane) by a Hohmann transfer to a circular drift orbit $a_d = a_t \mp h$ below (or above) target t. The chaser then dwells until the secular nodal gap closes, and finally makes a combined Hohmann and inclination transfer to t, plus a fixed proximity-operations allowance (15 m/s, stated). The depths considered are $h \in \{25, 35, 50, 75, 100, 150, 200, 300, 400, 500, 600, 700\}$ km (altitude $\geq$ 200 km), with dwell $\leq$ 2200 d and relative rate $\geq 0.01°$/day. After a leg the chaser lies in the target's plane, so its node depends only on (object, epoch). The tour is therefore a *time-dependent orienteering problem*: maximise the number of objects visited subject to a ΔV budget and a mission-time cap, where the arc cost depends on the departure epoch through the evolving gaps. Two envelopes are taken from the draft: "small" (2 km/s, 7 yr) and "large" (5 km/s, 15 yr). The counts include the free start object, since launch is assumed to insert the chaser at the start object. We report distributions over *all* start choices rather than the best of N.

2.4 Solvers and the scalarisation weight

Greedy and beam search rank actions by $w_t\,\Delta t/100\,\mathrm{d} + w_{\Delta V}\,\Delta V/100\,\mathrm{m/s}$. The draft used $w_{\Delta V}{:}w_t = 10{:}1$ and never tuned it. With both budgets binding, we find the Lagrangian-consistent weight by a sweep on random starts and report it on disjoint held-out starts (Section 6). We also test a rollout beam and a permutation NSGA-II (three seeds). The exact reference is a Held–Karp dynamic program over (visited set, current object) with Pareto labels (time, ΔV). It is exact for this leg model up to the weak departure-epoch dependence of drag make-up, and it is checked by brute force on tiny instances (`tests/`).

2.5 Baselines

All baselines use the same pool, leg costs and budgets. The metrics are objects reached, ΔV, time, and propellant mass by the rocket equation.

- *Impulsive*: direct plane-change hops, $2v\sin(\vartheta/2)$ plus Hohmann plus 15 m/s, hop time 2 d, beam width 6.
- *Natural drift only*: no altitude change; the chaser keeps its own orbit while waiting.
- *Published-style drift phasing*: one fixed raised waiting orbit at +40 km (the single raised orbit of [66]), and one fixed waiting orbit at −100 km (the carrier below the targets of [13]).
- *Free per-leg depth, lowering only* (this study).
- *Free per-leg depth, raising or lowering*, with one extra synodic lap allowed (the intermediate drift orbit "below, between or above" of [9]).

2.6 Uncertainty and verification

Sampled legs of the best tour are propagated numerically (Cowell, J_2–J_4, drag, solar radiation pressure, Sun and Moon) with SGP4 targets, and the node closure error is measured. A Monte Carlo then draws closure errors, calibrated on that check, with element-set growth, burn execution, density and lognormal proximity-operations variability, and counts the objects completed within the budget. A one-at-a-time sensitivity study uses paired differences over fixed starts.

2.7 Collision-risk benefit

A kinetic-gas model with latitude-resolved spatial density gives the catastrophic-collision involvement rate of each object against the catalogue population. The cross-section is the sum of radii, a collision is catastrophic if $\frac{1}{2}m_s v^2/m_l \geq 40\,\mathrm{J/g}$, and mass is obtained from the radar cross-section (RCS) by an imputed relation, with 66% of radar cross-sections imputed. Fragment yield follows the standard breakup power law [37]. The model is a ranking and scaling tool with large uncertainty and is *not* a prediction; the draft's superseded "95.4%" claim is not re-derived.

2.8 Key assumptions

The main modelling assumptions are summarised below; their effect is discussed in Section 13.3.

- Legs are impulsive and circular-to-circular, with no in-plane phasing; proximity operations and capture are one fixed allowance of 15 m/s.
- Launch inserts the chaser at the start object at no cost.
- Drag make-up uses a tabulated exponential atmosphere and $C_d A/m = 0.01\,\mathrm{m^2/kg}$.
- The dry masses and specific impulses of the two propulsion designs, and the kit mass, are stated assumptions rather than sourced values.
- The node uncertainty at epoch (0.01°) is an assumption, since no published RAAN uncertainty was found.

3 Audit of the Draft Study

Table 1 lists the findings that change a headline or a stated claim; the full list (13 analytic and 13 pipeline items) is in `audit/AUDIT.md`. The independent engine reproduces the draft's own tour (29 objects, 1,445.48 m/s, 2,555.54 d, node closure 2×10^{-13} deg at all 28 legs), so the differences between the two studies are model or input choices.

3.1 Reproduction of the 2025 pipeline

An earlier version of this audit could not reproduce the draft's catalogue counts, because the 2025 inputs were not in the repository. With the original files recovered, we ported the two 2025 stages to Python and ran them inside the work folder: the MATLAB cluster locator, which groups non-operational payloads and rocket bodies at 800–1,100 km by a greedy 1° inclination rule, and the Python discovery module (DBSCAN on inclination with $\varepsilon = 0.5°$, then sliding windows of 10–20 objects with altitude, eccentricity and RAAN range limits). Table 2 gives every stage.

Three facts explain the draft's numbers. First, "63,553 rows" is the line count, header included, of the April 2025 catalogue (63,552 objects: 33,505 re-entered and 30,047 in orbit, of which 14,699 payloads, 12,916 debris, 2,368 rocket bodies and 64 unknown). Second, the 1,288 actionable objects come from a different, earlier file, the March 2025 catalogue that the 2025 code actually

Table 1: Audit Findings That Change a Headline (Severity A) or a Stated Claim (B)

	Draft claim	Finding
A	63,553 rows → 1,288 actionable objects; 40 sub-clusters, 593 objects	Reproduced exactly from the 2025 files (Section 3.1), but the three counts come from two different catalogues and include decayed objects: 193 of the 1,288 and 63 of the 593 had re-entered. On the 2026 snapshot the rule restricted to objects in orbit gives 1,151 and 913 in 51 sub-clusters
A	Hohmann cost symmetric; net advantage $1+4.5x$	$\Delta V_{\downarrow}/\Delta V_{\uparrow} = 1 + \frac{3}{2}x + \frac{9}{8}x^2$; net per-$\Delta V$ gain $1+3x$
A	Efficiency has an interior optimum 150–300 km below	No interior optimum (gain per ΔV rises monotonically); quoted values about 11 times too large
A	"59.8% saving" over plane changes	No time-free saving exists. At 1° the saving is 80% if 279 d are allowed, 13% if 30 d. Same-sequence replacement on the draft tour: 72.5%
A	Headline counts (24, 29, 53 objects)	Best-of-60 starts; over all 159 starts the draft's configuration has median 20 (CI 19–20), maximum 28
A	Weights 10:1	Far from the Lagrangian optimum; counts understate the same model by 45–90%
A	"Objects removed"	The tours *visit*; removal needs one kit per object (Section 9)
B	Alignment window 1°	Defined by cost: 12.8 m/s per 0.1°; the 15 m/s allowance covers 0.118°
B	$R > 1$ "for every target"	False for eccentric targets (reversal near $e \approx 0.03$)
B	Greedy and differential evolution	Results come from greedy, beam and NSGA-II; DE exists in the repository but is not behind the numbers
B	Elements	Osculating inputs: nodal-rate noise 0.0026°/day (1σ) against a pool spread of 0.0066°/day

read, and the rule has no decay filter, so 193 of them had already re-entered; on the April file the same rule gives 1,297. Third, the 40 sub-clusters with 593 objects require the RAAN join of 2025: the sub-cluster windows reject RAAN ranges above 180°, so the result depends on which objects have a node angle. Joining the April 2025 element sets (11,806 sets) and seven cluster exports (389 records, 24 of them with epochs as early as 1984, i.e. the last elements of re-entered objects) gives 662 of the 1,288 objects a RAAN and reproduces the shipped catalogue exactly: the same 593 objects in the same 40 sub-clusters and 8 parent clusters, with 250 of the 270 shipped RAAN values equal to 10^{-4} deg (the other 20 differ by 0.01–0.13°, consistent with averaging over a slightly different set of duplicate element sets). The MATLAB locator output (527 objects in 42 groups) is also reproduced with identical labels. That the DBSCAN step also yields 40 inclination clusters is a coincidence. The numbers in the rest of this paper are unchanged, because no tour uses the sub-cluster catalogue.

The 2025 MATLAB optimiser. The 2025 MATLAB tours differ from the Python results for reasons of model, not of solver quality. The MATLAB optimiser tours one preselected sub-cluster of 20 objects (or one RAAN group of about 25), so its count is capped by the set; it starts at the largest RAAN and admits only targets with a smaller node, with no wrap-around; it ranks

Table 2: Catalogue Stage Counts of the 2025 Pipeline, Reproduced, and the Same Rules on the 2026 Snapshot

Stage	Mar 2025	Apr 2025	Aug 2026
catalogue rows (newest launch)	63,203 (21 Mar)	63,552 (27 Apr)	70,292 (17 Aug)
of which re-entered	33,032	33,505	35,473
numeric apogee, perigee and inclination	62,297	62,642	68,244
mean altitude 0–2,000 km	56,304	56,646	62,272
non-operational rule (R/B, or PAY with status "−")	5,582	5,619	5,736
of which at 600–1,500 km ("actionable")	**1,288**	1,297	1,349
of which re-entered	193	194	198
in orbit	1,095	1,103	1,151
DBSCAN inclination clusters (noise)	40 (13)	39 (14)	40 (19)
sub-clusters / objects, 2025 RAAN joined	**40 / 593**	36 / 552	–
sub-clusters / objects, no RAAN	51 / 900	–	53 / 952
MATLAB locator, 800–1,100 km: objects / groups	527 / 42	530 / 43	–

Rules exactly as in the 2025 code; no decay filter except in the "in orbit" row. The v3 pool of this paper (1,151; 51 sub-clusters, 913 objects) applies the decay filter before clustering. The 2026 sub-cluster counts without RAAN include decayed objects. Source: `analysis/a9_orig2025.py`.

targets with the drift rate of a fixed reference orbit ($a = 7{,}100$ km) and then optimises the per-leg depth (50–400 km) by particle swarm; it uses RAAN values from element sets of mixed epochs without propagation; and it charges no inclination difference, no proximity-operations allowance and no drag. Its SL-8 tour of 20 objects (19 legs, 2,000 m/s, 511 d) and its sun-synchronous tours (26 objects in 4,553 d, or 21 in 1,671 d, each at 2,000 m/s) are therefore not comparable with the 29 (SL-8) and 27 (sun-synchronous) medians of Table 4, although the sun-synchronous figures are of the same order. Several exploratory 2025 outputs carry sign errors (negative or 10^5-day total waits) and are not used. Of the 2025 graphics, the three-panel chart of wait time against forward RAAN difference (exported as a 1.2 s clip) is worth showing as a historical record of the first tours, with the caveat on unpropagated epochs; the 94 s trajectory animation (840×630 pixels) mislabels metres as kilometres and is not.

4 Lowering versus Raising: Exact Results

We first establish the exact comparison between waiting below and waiting above a target. Let $x = \Delta a/a_T \in (0, 1)$ and $f(a) = a^{-7/2}$.

Theorem 1 (Rate gain)**.** *For circular orbits with the same inclination and J_2 only, $R(x) \equiv \dfrac{|\Delta\dot\Omega_\downarrow|}{|\Delta\dot\Omega_\uparrow|} = \dfrac{(1-x)^{-7/2}-1}{1-(1+x)^{-7/2}} > 1$ for every $x \in (0, 1)$, independent of i.*

Proof. f is strictly convex, so $f(a-\Delta)+f(a+\Delta) > 2f(a)$, i.e. $f(a-\Delta)-f(a) > f(a)-f(a+\Delta) > 0$. The factor $K\cos i$ cancels. □

Proposition 1 (Series and cost)**.** *$R = 1 + \frac{9}{2}x + \frac{81}{8}x^2 + \frac{141}{8}x^3 + O(x^4)$, with $0 \le R - 1 - \frac{9}{2}x \le 13.58\,x^2$ for $x \le 0.15$ (supremum checked on a 20,000-point grid and symbolically). The exact Hohmann costs satisfy $\Delta V_\downarrow = \frac{1}{2}x + \frac{3}{8}x^2 + \dots$ and $\Delta V_\uparrow = \frac{1}{2}x - \frac{3}{8}x^2 + \dots$ (units of v_c), hence $\Delta V_\downarrow/\Delta V_\uparrow = 1 + \frac{3}{2}x + \frac{9}{8}x^2 + O(x^3)$, and the gain per unit ΔV is larger for lowering by $R\,\Delta V_\uparrow/\Delta V_\downarrow = 1 + 3x + O(x^2)$.*

At $a_T = 7350$ km (975 km altitude), $R = 1.031, 1.063, 1.130$ and the equal-ΔV gain ratio is $1.021, 1.042, 1.087$ for offsets of 50, 100 and 200 km (first-order $1 + 3x$: 1.020, 1.041, 1.082). The error order $O(x^3)$ stated in the draft is therefore wrong, since the remainder is $O(x^2)$. Figure 1 shows these results.

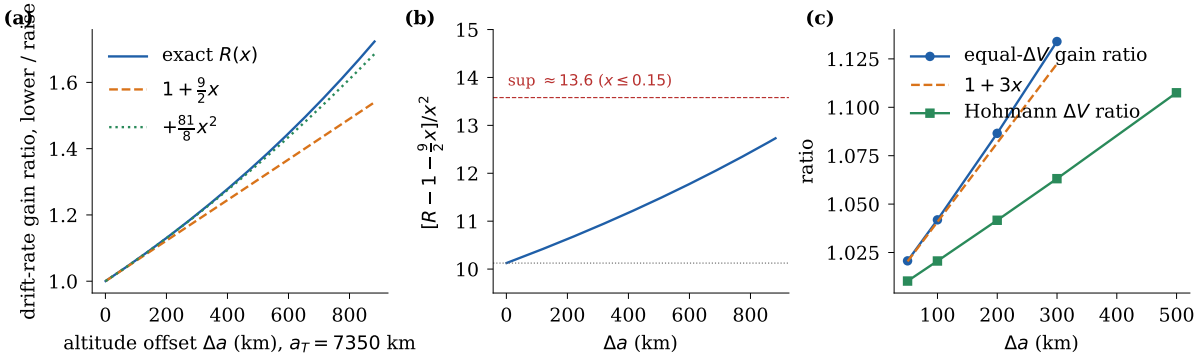


Figure 1: Lowering versus raising for circular orbits. (a) Exact rate-gain ratio $R(x)$ and its series. (b) Scaled remainder, bounded by 13.58. (c) Hohmann cost ratio and equal-ΔV gain ratio against $1 + 3x$. Lowering wins in both rate and cost, but by only a few per cent at practical offsets.

Both signs, inclination and eccentricity. For retrograde orbits ($\cos i < 0$) the rates change sign and "lowering" accelerates the drift eastward; Theorem 1 is unchanged. Including J_2^2 and J_4 changes R by at most 1.1×10^{-3} across $i = 52$–$98.7°$. A circular chaser against an *eccentric target* sees an effective altitude offset, because $\dot{\Omega}_t \propto a^{-7/2}(1 - e^2)^{-2}$ is the rate of a circular orbit lowered by $\frac{4}{7}e^2 a$ (0.1 km at $e = 0.005$, 10.5 km at 0.05, 42 km at 0.1). For a 100 km offset, $R = 1.063, 1.054, 1.028, 0.861, 0.433$ at $e = 0, 0.01, 0.02, 0.05, 0.1$, so raising wins above $e \approx 0.03$. This is irrelevant to the SL-8 pool ($e \approx 0.002$) but affects 62 of the 593 shipped objects.

Drag. The $1+\frac{3}{2}x$ cost ratio applies to the transfer alone. When the dwell make-up is added, the lower-to-raise ΔV ratio at 975 km is 1.009 (50 km offset, mean activity); near 600 km it is 1.1 (mean) to 1.4–1.5 (high activity). The draft's "2 km/day drag floor" never binds (decay at 300 km altitude is 0.36 km/day); the effective limits are the 200 km altitude floor and the make-up cost (about 1 m/s per day at 275 km).

Decision rule. Lowering only closes gaps of one sign (westward for prograde orbits). For a gap $g \in (0, 360°)$ in the lowering direction, lowering is faster when $g < g^* = 360R/(1 + R) \approx 180°(1 + \frac{9}{4}x)$; the approximation errs by $\le 0.46°$ for $x \le 0.1$. Allowing both senses halves the expected wait for a uniform random gap (Monte Carlo ratio 1.94, theory 2), and on SL-8 pairs the rule picks the faster sense in 94–96% of cases (wait ratio 2.15 at 50 km). The "always lower" design rule is therefore a weakness rather than a result, and the bidirectional planner is the right baseline.

5 Validation of the Numerical Model

Secular rate. Against a J_2-only Cowell integration, Eq. (1) evaluated on mean elements is within 0.04–0.08%; with the osculating a it is 0.34–0.46% off. Adding J_3 and J_4 shifts the rate by 0.02–0.23%, an effect dominated by J_4 (Fig. 2).

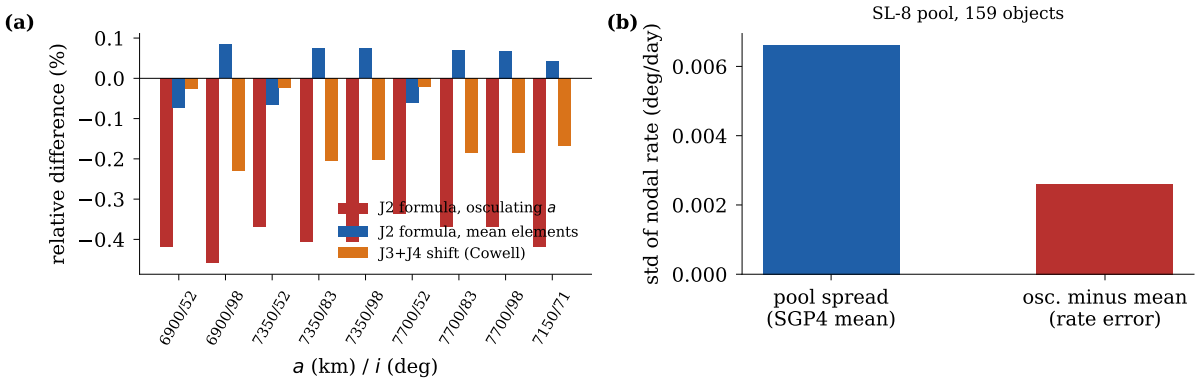


Figure 2: Validation of the secular nodal rate. (a) Secular-formula error against Cowell integration for nine (a, i) cases. (b) Pool spread of the nodal rate against the rate error from using osculating elements; the osculating error is a significant fraction of the spread, which is why mean elements are used.

Leg closure. For eight legs of the tuned best tour, a circular Cowell chaser (full forces) and SGP4 targets give node closure errors of $-0.002 \pm 0.017°$ (mean $\pm$ s.d., maximum $0.039°$) for the first-order plan and $-0.006 \pm 0.018°$ (maximum $0.049°$) for the second-order plan. These correspond to about 1.3–1.5 m/s of correction per leg, against the 15 m/s allowed ($0.118°$). A J_2-only numerical model disagrees by $0.08°$ because the plan includes J_4; adding J_3 and J_4 closes this to $0.003°$, and drag, SRP and Sun/Moon account for the rest (Fig. 3). Planned and propagated dwell differ by up to 3 d. Eight legs is a small sample, and Section 13.3 states what is not covered.

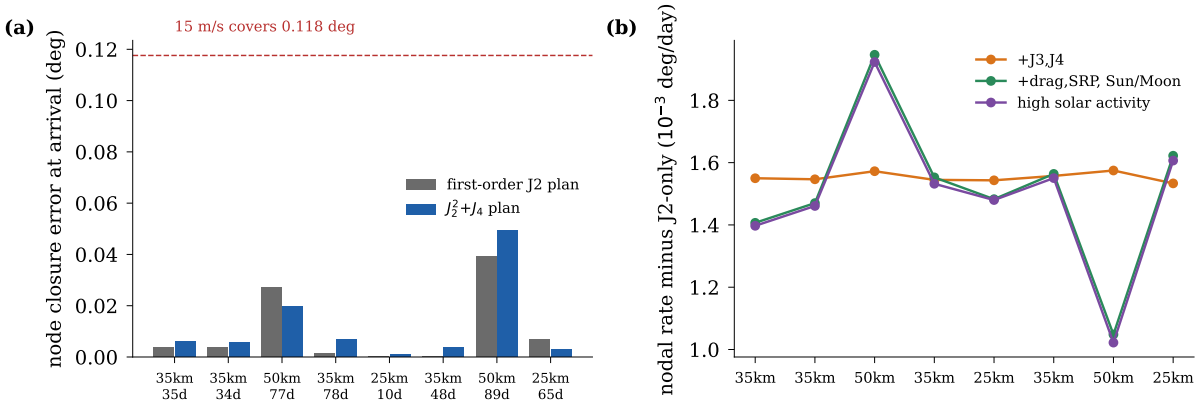


Figure 3: High-fidelity verification of the leg model. (a) Node closure error at arrival on eight sampled legs; the dashed line is the error covered by the 15 m/s allowance. (b) Nodal-rate increment of each added force model over J_2 only.

Code. Twenty automated tests pass. They cover the zonal accelerations against the potential gradient; the series and bound of Proposition 1; the Hohmann ratio and the plane-change small-angle limit; the second-order rate against SGP4; the drag identity; the replay of the draft's tour; the closure and budget invariants; the dynamic program against brute force; the funnel counts; the risk-model scaling; the SMAD learning-curve, pass-duration and cost-uncertainty relations; and the exact reproduction of the 2025 counts.

6 Optimiser Validation

Weight. We swept $w_{\Delta V}{:}w_t$ on random starts and evaluated the result on disjoint held-out starts. The best weights are 2 (small envelope) and 1.5 (large), against 10 in the draft, and the held-out medians are 31 versus 22 objects (small) and 68.5 versus 39.5 (large). Figure 4 shows the sweep. This is standard multiplier tuning, not a new technique; the point is that the draft's published counts understate its own model.

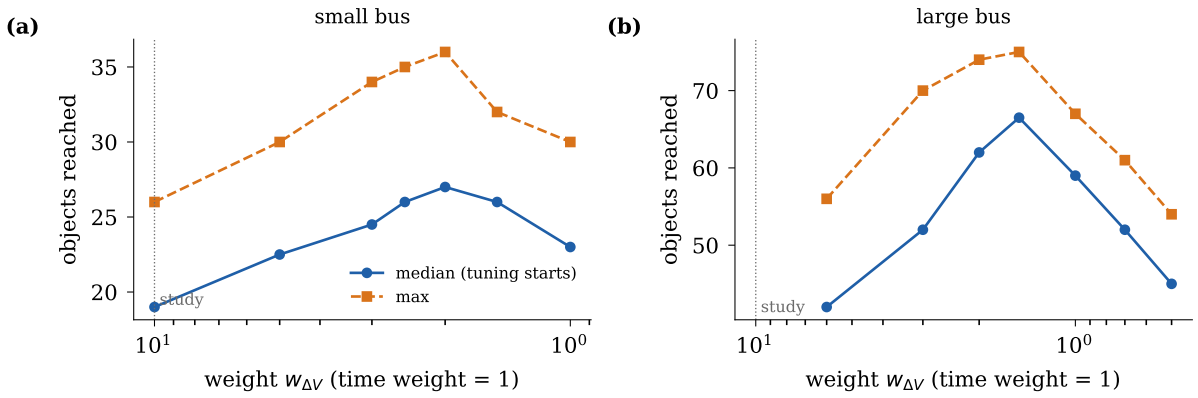


Figure 4: Objects reached against the scalarisation weight. The draft used 10, well away from the best values for either envelope.

Distance from optimum. On 32 small instances (16 windows of 10 and of 12 consecutive-RAAN objects, with budgets of 700 m/s and 600 d, and of 800 m/s and 700 d) the exact dynamic program gives the optimum count. The mean gaps (95% bootstrap CI) are: greedy at the draft weight 18.7% (16.0–21.4); beam-20 at the draft weight 10.3% (8.0–12.7); NSGA-II 9.7% (7.1–12.2); tuned greedy 6.1% (4.1–8.2); tuned beam-20 1.9% (0.6–3.4), optimal in 81% of instances (worst 12.5%); and rollout beam 2.6% (1.0–4.2). Beam width saturates at about 10 on the full pool (best start 34, 34, 35, 35, 35, 35, 35 objects for widths 1–1000; Fig. 5). The ΔV-only relaxation bound is on average 31% above the exact optimum on the small instances. On the full pool it is 71 objects at 2 km/s (best found 36) and vacuous at 5 km/s (160 against a pool of 159), so the full-pool gap is not certified and the small-instance gaps are the evidence. The draft's belief that wider beams or evolutionary search would help is not supported; the weight mattered more.

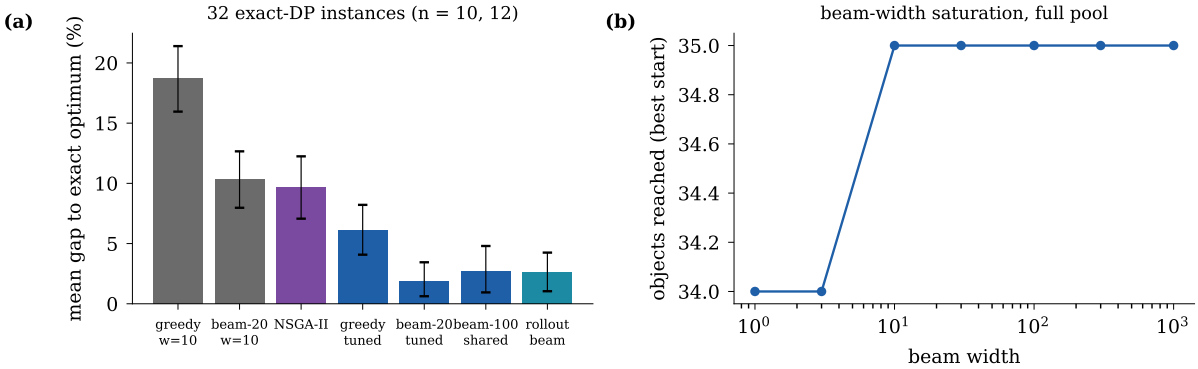


Figure 5: Optimiser performance. (a) Mean gap to the exact optimum with 95% CIs. (b) Beam-width saturation on the full pool. The tuned beam search is close to optimal, and widening the beam beyond about 10 gives no gain.

7 Results

7.1 Fair comparison with natural drift and impulsive plane changes

To place the drift tours in context, we compare them with every baseline at equal budget. Table 3 gives the medians over all 159 starts (bootstrap CI), with ΔV, time and propellant, and Figure 6 shows the full distributions. Three comparisons from Table 3 warrant discussion. First, at equal

Table 3: Objects Reached (Including Start), SL-8 Pool, All 159 Starts

Method	median	95% CI	max	ΔV (m/s)	time (d)	chem (kg)	EP (kg)
small envelope: 2 km/s, 7 yr							
draft as audited	20	19–20	28	1664	2555	489	89
v3 inputs, draft weight	20	19–22	27	1707	2554	506	91
fixed raise +40 km	16	14–17	30	851	2451	218	44
fixed lower −100 km	18	18–18	18	1977	1010	614	107
free lower, tuned	29	27–31	36	1968	2513	611	106
free bidirectional + laps	31	28–32	36	1971	2479	612	106
natural drift only	6	6–7	12	114	2267	26	6
impulsive plane change	9	8–10	15	1775	16	532	95
large envelope: 5 km/s, 15 yr							
draft as audited	35	32–38	52	2500	5477	853	137
fixed raise +40 km	32	29–36	51	1779	5329	534	95
fixed lower −100 km	44	44–44	44	4998	2732	2742	295
free lower, tuned	67	64–68	76	4976	5325	2718	293
free bidirectional + laps	67	67–70	77	4974	5342	2716	293
natural drift only	12	12–13	22	440	5138	105	23
impulsive plane change	19	18–22	31	4779	36	2510	280

Propellant (fixed-orbit rows: rocket equation on the median ΔV): storable chemical design (700 kg dry) and electric design (900 kg dry, 1800 s), rocket equation.

ΔV budget the drift tour reaches 3.2 times (small) and 3.5 times (large) as many objects as impulsive hopping, but it needs 2,500–5,300 d against 16–36 d. The saving is a trade of time for propellant, and the value of that trade depends on the bus life, not on a universal percentage. Second, waiting with no altitude change is cheap (114 m/s) but slow, reaching only 6 objects. Third, the published-style *fixed* waiting orbits do worse than free per-leg depth: the +40 km orbit reaches 16 (median) and the fixed −100 km orbit 18, against 29. The −100 km result is deterministic here, since every leg costs nearly the same and the budget binds. The gain over the published rules therefore comes from the per-leg choice of depth and sign. The bidirectional planner adds about two objects at 2 km/s and nothing measurable at 5 km/s.

7.2 Headline missions on the frozen snapshot

Table 4 gives the headline re-run of the two envelopes with the tuned weights. The draft's 29 is reproduced exactly by the independent engine with the draft's settings, and it is the maximum over starts of a configuration whose median is 20. With the weight corrected, the median is 29 and the best 36. In the large envelope the pool is nearly exhausted by the best tours (76 of 159), so the counts there are pool-limited and are not a measure of the method's reach. Figure 7 shows the two best tours.

7.3 Collision-risk benefit

To connect the tours to the debris environment, we compute the catastrophic-collision involvement rate removed by each median tour (Table 5, Fig. 8). For the SL-8 median tour the removed hazard

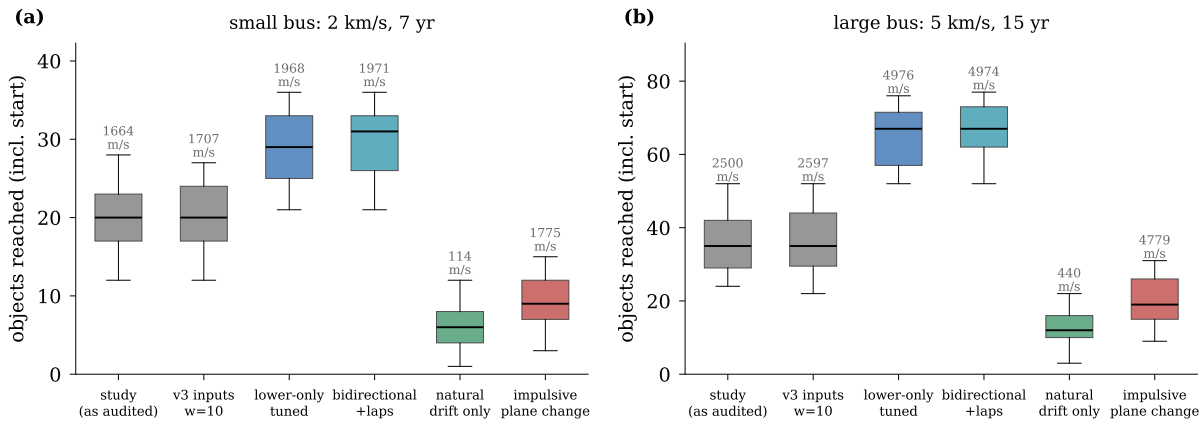


Figure 6: Objects reached over all 159 starts for each method (box: quartiles; whiskers: range); labels give the median ΔV. Free per-leg depth reaches the most objects, while impulsive hopping is the only fast option.

Table 4: Headline Re-run on the Frozen Snapshot

	2 km/s, 7 yr	5 km/s, 15 yr
tuned weight	2	1.5
median over all starts (95% CI)	29 (27–30)	67 (64–68)
greedy median	28	65
best start	36 (1995 m/s, 2397 d)	76 (4981 m/s, 5090 d)
best-of-60 starts, mean (CI)	35.3 (34–36)	75.9 (75–76)
Monte Carlo, best tour: refreshed	33.0	71.8
unrefreshed (open loop)	28.7	58.9
pessimistic errors	30.3	67.4
draft as published (best of 60)	29 (1445.5 m/s, 2555.5 d)	53 (4101 m/s)
four bands (SL-8, SSO, Iridium, SL-16), medians summed	87 (best 101)	171 (best 188)

SL-8 pool, beam width 12 on all 159 starts unless stated; Monte Carlo of execution errors on the best tour.

is 18% of the pool's total (0.0088 events/yr). Over 25 years the model gives 0.040 catastrophic collisions avoided (Monte Carlo 5–95%: 0.025–0.055) and about 65 fragments > 10 cm (21–217), while a random set of the same size gives 0.039. In SL-8 the per-object hazard is nearly uniform (CV 0.16), so *risk-aware selection is worth nothing* there. In the sun-synchronous band it would double the benefit (CV 0.67), but the tour as sequenced captures only half of what the top-n set would. Choosing the start by risk raises SL-8 from 28 to 36 objects and the benefit from 0.040 to 0.050. The cost per avoided event is a scaling statement, not an estimate: at an assumed \$100–400 M per mission, the nominal figure is \$2.5–10 B per avoided catastrophic collision (\$1.8–16 B across the 5–95% interval of avoided events). Published cost anchors are sparse and are not used as inputs. The result must not be read as a case for or against removal, because the model has no fragment-cascade feedback beyond the first generation (amplification 1.06) and the long-term benefit of avoiding the *first* fragmentation in a band is not captured.

8 Uncertainty and Sensitivity

Execution errors. The closure-error Monte Carlo (calibrated on Section 5) yields, for the planned 35-object tour at the 2 km/s budget, 32.0 objects completed with an element refresh at each leg (a receding-horizon replan), 27.8 without refresh (the 95th percentile node error is then 0.36°, three times the 15 m/s allowance) and 28.8 under pessimistic errors. A planning margin of 0–15% of the budget does not help the open-loop case (27.5–27.9 at every margin), so the

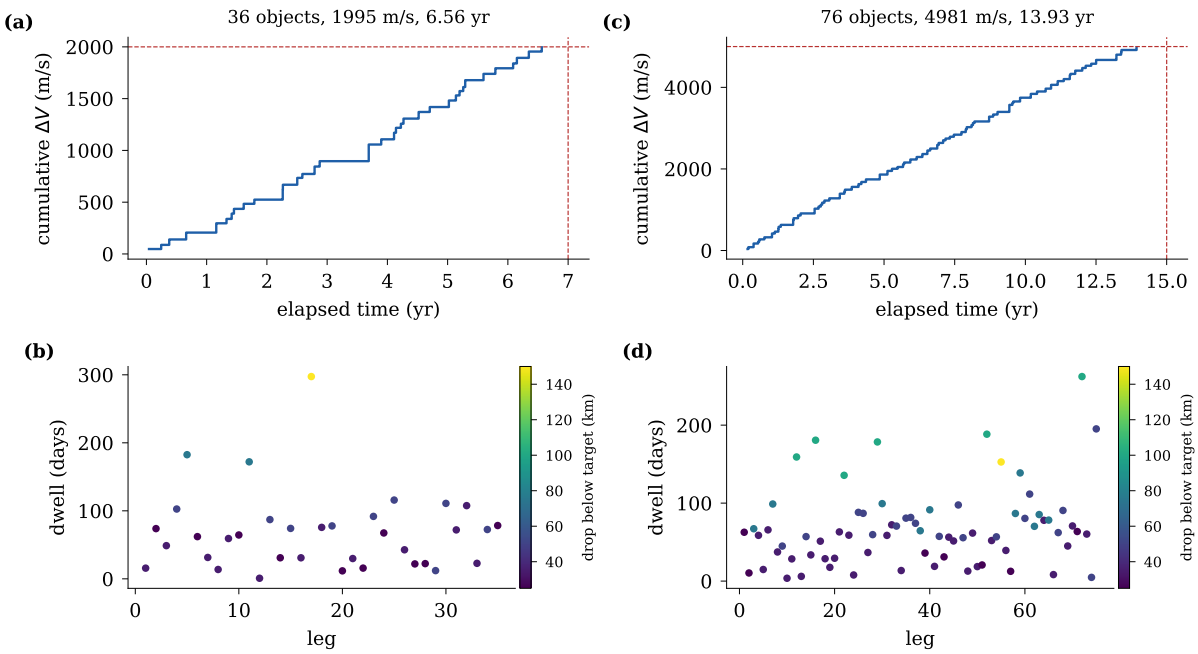


Figure 7: Best tours for the two envelopes. (a,c) Cumulative ΔV against time; (b,d) dwell per leg, coloured by depth below the target.

Table 5: Catastrophic-Collision Involvement Rate Removed by the Median Tour (Kinetic-Gas Model; *Ranking-Level Accuracy Only*)

band	pool	tour n	tour rate (10^{-3}/yr)	top-n	random	CV
SL-8	159	29	1.58	1.85	1.61	0.16
SSO	135	27	1.27	2.69	1.31	0.67
Iridium-class	28	11	0.55	0.65	0.53	0.27
SL-16	20	16	0.87	0.89	0.87	0.10

Tour n is the median object count including the free start; the benefit calculation below uses the median-hazard tour (28 objects for SL-8). CV: coefficient of variation of per-object rate; "top-n": removing the n highest-rate objects; "random": random n.

recovery comes from refreshing the elements and not from reserve (Fig. 9). The first-order draft model and the second-order model give nearly the same count (32.1 and 32.0), because the rate bias is common mode.

One-at-a-time sensitivity. The sensitivity study uses paired differences over fixed starts (baseline 28.1 objects, Fig. 10). The largest losses come from a relative-rate floor of 0.03°/day (−9.8, CI −11.1 to −8.5); the draft weight of 10 (−8.9); a proximity-operations allowance of 50 m/s (−8.2) or 30 m/s (−3.7); an in-plane phasing allowance of 30 m/s and 20 d per leg (−7.3); a 5 yr life (−5.6); and a 1.5 km/s budget (−5.5). The gains come from a 3 km/s budget (+2.6), a 10 yr life (+1.8) and a 5 m/s allowance (+1.6). The ballistic parameter (0.003–0.03), solar activity, the decay cap, an altitude floor of 200 km, the leg-wait cap and inclination charging have exactly zero effect. The drag insensitivity is a property of this pool (drops of 25–150 km below 975 km, where decay is negligible) rather than a general result, and it would not hold for bands near 600 km. In-plane phasing is not otherwise modelled (closing 90–180° over 20 d costs 4.5–9 m/s round trip, and 18–36 m/s over 5 d); this is the most important unmodelled cost.

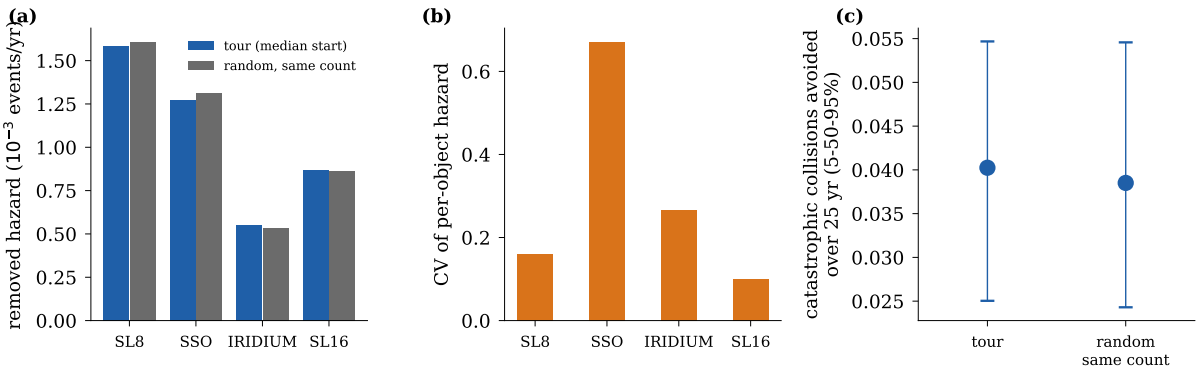


Figure 8: Collision-risk benefit of the median tours. (a) Removed hazard of the median tour and of a random set of equal size. (b) Heterogeneity of per-object hazard. (c) Collisions avoided over 25 yr. Ranking-level accuracy only.

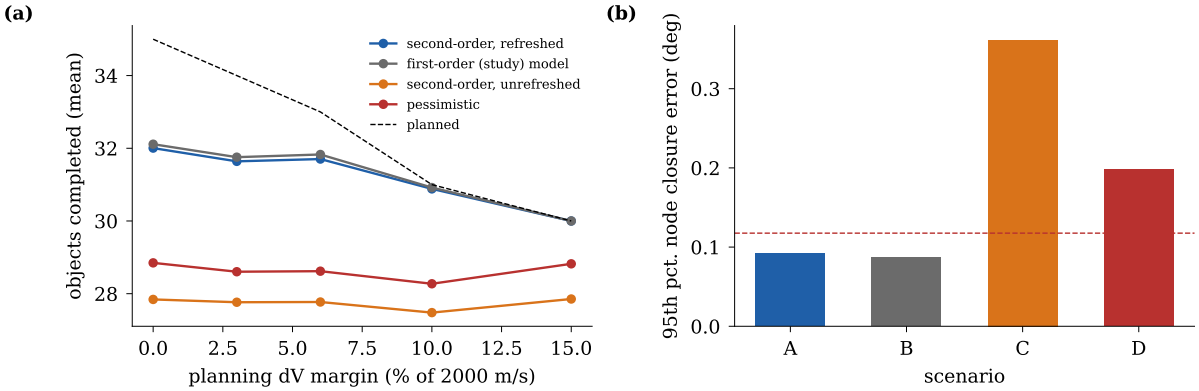


Figure 9: Monte Carlo of execution errors. (a) Objects completed against planning margin for four scenarios. (b) 95th percentile closure error (A: second-order refreshed; B: first-order draft; C: unrefreshed; D: pessimistic; dashed: error covered by the allowance). Refreshing the elements, not adding margin, recovers the lost objects.

Beyond secular J_2. At 83° the absolute luni-solar nodal rate is 6×10^{-5}°/day, and its *differential* across a 100 km drop is 1.2×10^{-6}°/day, or 3×10^{-5} of the J_2 differential. With drag, SRP and third bodies, the full Cowell model differs from J_2–J_4 by 1×10^{-4}–4×10^{-4}°/day (below 0.05%). The SGP4 element-set error is the dominant data uncertainty: the TLE range error grows by about 1.5 km/day [44], no published RAAN uncertainty was found, and the node uncertainty at epoch (0.01°) is an assumption.

9 Novel Methods

Each idea is simulated against the baselines of Table 3 on the SL-8 pool with fixed seeds; the objective is the number of objects reached or placed, ΔV, time, and launch or propellant mass. The verdicts use only the numbers, and "prior art" cites what Section 13.2 found. Table 6 summarises the screen and Figure 11 shows the carrier comparison.

The one supported idea. A carrier parks on a *single persistent* lowered drift orbit (depth 400 km, $\Delta V \approx 210$ m/s including make-up) whose plane sweeps through the cluster. When the node of its orbit coincides with that of a target within the tolerance, a kit departs (kit ΔV 260 m/s, 5 days, 30 m/s extra), and the carrier does not return to any target plane. The K

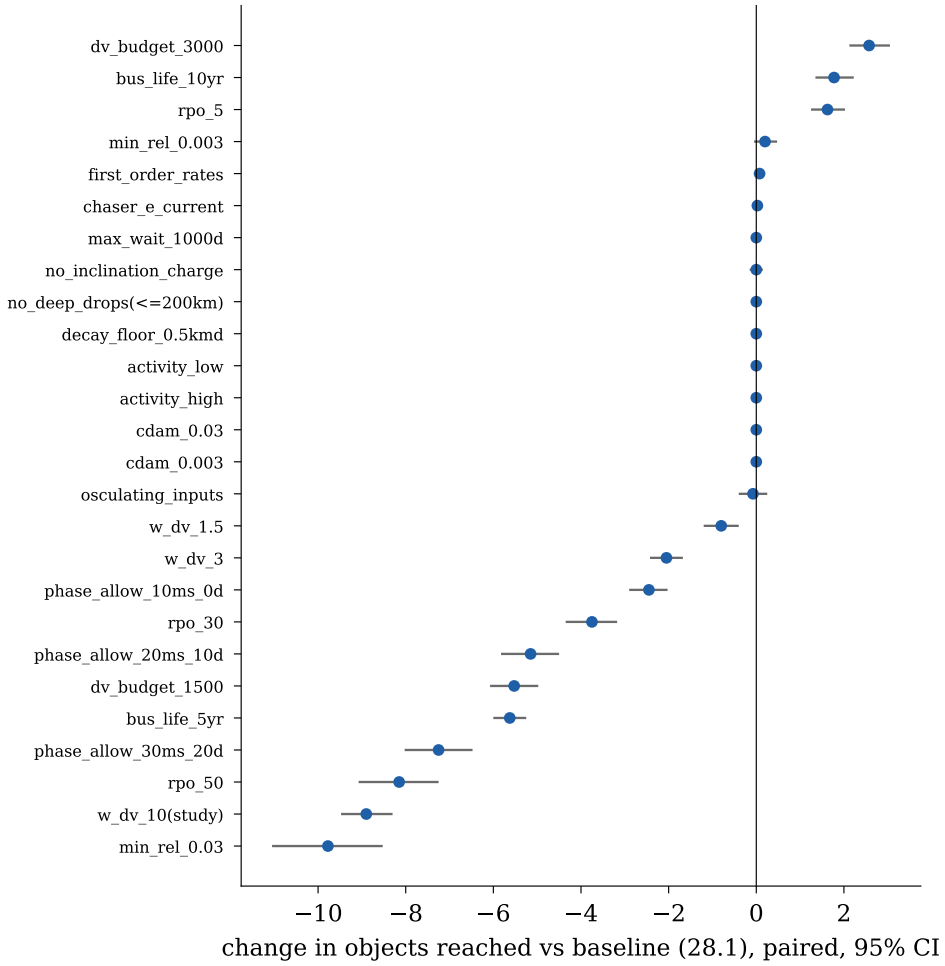


Figure 10: One-at-a-time sensitivity: change in objects reached against the baseline, paired, with 95% CI. The proximity and phasing allowances dominate; drag has no effect for this pool.

nearest-in-time targets are served in the order in which the sweep reaches them. Depth trades time against launch mass (Fig. 11), and the time CI is a bootstrap over all 159 starts. The comparison fixes the kit mass at 135 kg (unverified), counts kit propellant with $I_{sp} = 280$ s, and omits kit avionics, capture hardware and rendezvous autonomy. The launch-mass advantage (5% at $K = 8$, 20% at $K = 16$, 50% at $K = 36$) is therefore an upper bound. The time advantage (4.5 to 6.3 times) is less sensitive: kit allowances of 60–100 m/s and 10–20 d change the median at $K = 16$ from 298 to 313 d at 300 km depth. The hopper medians use 20 sampled starts. Prior art covers a carrier with kits that waits in a drift orbit [11, 13, 66]; whether kits depart from a persistent orbit without a carrier return was not found in the sources reached, but absence in a bounded search is not proof.

10 Servicer Design and Comparison with Demonstrations

The tours above count visits by a point mass. This section asks what spacecraft would fly them, following the subsystem-level process of SMAD [43]. Unless stated, the reference is the supported architecture of Section 9: a chemical carrier parked 400 km below the SL-8 cluster that releases $K = 16$ kits (the median start, Table 6), against a serial hopper that carries the same kits, and a visit-only hopper on the 2 km/s, 7 yr envelope. Inputs that are assumptions are tagged (A) in Appendix B; the computations are in `analysis/a8_servicer.py`.

Table 6: Screen of Ten Ideas plus the High-Fidelity Check (EP: Electric Propulsion)

Idea	Result	Cost	Verdict
Continuous low-thrust profiles	Objects 25.5–28 (CI 24–31) against 28 impulsive; propellant 97–105 kg against 610 kg, entirely from I_{sp} (1800 s against 320 s)	thruster, power, time	no trajectory gain; prior art [9, 38]
Eccentric drift orbit / apsidal	Gain per ΔV 0.970–0.998 of circular, falling with depth	none	rejected
Oberth-type plane change at high apogee	Direct plane change cheaper than the bielliptic option for gaps up to 30° (127.5 against 163 m/s at 1°); drift 25–55 m/s for 1–10° in 2 yr	–	rejected for small angles
Luni-solar plus J_2	Differential rate 3×10^{-5} of J_2	–	rejected
Drag sail as altitude shaper	Time to descend 100 km for C_dA/m = 1–0.1 m^2/kg, mean activity: 16–156 d at 500 km, 77–767 d at 600 km, 330–3,270 d at 700 km, 840–8,370 d at 775 km, 4,700–47,000 d at 975 km (high activity 5–9 times faster)	sail area and mass; no control of re-raise	useful below about 700 km; impractical at 975 km
Deorbit of removed object to retime	No mechanism: the removed object leaves the pool, the chaser's orbit is unchanged. Towing a 1,435 kg object 100 km down costs 50.6 m/s, triples the propelled mass, and does not deorbit it at 975 km	–	rejected
Better tour formulations	Tuned weight: +9 objects (small), +32 (large) median; rollout beam and exact-gap results (Section 6); receding-horizon refresh +4.2 objects (Section 8)	compute only	supported, but standard
Two or three servicers, one cluster	Same total ΔV split 1/2/3 ways: 35/36/39 objects, time 2493/1380/887 d, propellant 621/513/477 kg; k full servicers: 35/65/94	more hardware	time and mass benefit, no count benefit
Band choice by RAAN concentration	40-object pools: RAAN-window 21, random 16, evenly spread 14 objects	–	supported (pool design)
Parked carrier + expendable kits	K = 16: placement 219 d (CI 195–246) against 1198 d for a hopper; launch mass 3.3 t against 4.1 t; kit $\Delta V \leq 261$ m/s; K = 36: 484 d against 3041 d, 6.5 t against 13.4 t	kit autonomy and kit bus mass (not counted)	**candidate**; partial prior art
(verification) Full Cowell model	Full Cowell against J_2–J_4: slope differences $\leq 4 \times 10^{-4}$°/day	–	supports the model

10.1 Where the drift-based tour fits

We first place the concept against the servicing and removal missions that have flown or are funded (Table 7, Fig. 12). Masses and costs are as published by the agencies or in the NASA servicing survey [4]; values found only in databases or trade reports are marked, and no mission in the sources reached publishes its ΔV capability.

Table 7 shows a clear pattern. The demonstrations retire the risk of one step at a time: capture of a prepared or own client (RemoveDEBRIS, ELSA-d), non-cooperative rendezvous with a rocket body to about 15 m (ADRAS-J, reported in a secondary source), and capture of one unprepared object (ClearSpace-1, ADRAS-J2). In the terms of SMAD Table 20-15 [43, p. 804], rendezvous with a non-cooperative rocket body has been tested in space, while capture of a tumbling, unprepared rocket body has not, so it sits at about TRL 4–5 (our reading). The drift-phased tour does not compete with these missions; it is the step after them. Its advantage is the number of objects per launch, which no demonstration addresses, and its cost is the years of drift and the need for many kits that each perform a capture. The GEO life-extension servicers, which move between clients, and the pods installed by a robotic vehicle on several clients [4] are the closest operational analogue of a carrier that distributes kits.

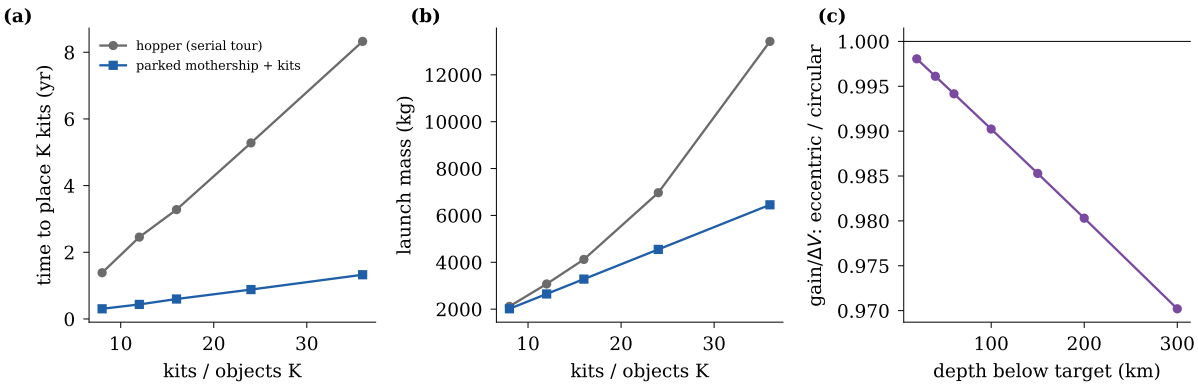


Figure 11: Novel-method comparisons. (a) Time to place K kits: serial hopper against a parked carrier at 400 km depth. (b) Launch mass. (c) Eccentric against circular drift orbit, gain per ΔV. The carrier places kits several times faster; the eccentric drift orbit gives no gain.

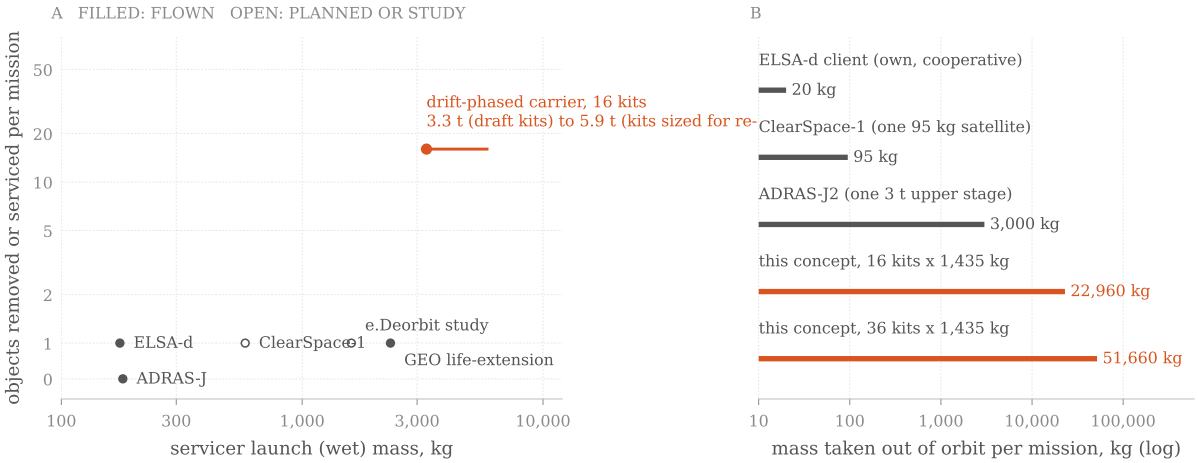


Figure 12: Servicer launch mass against objects per mission, and mass taken out of orbit per mission. Every flown or planned demonstration handles at most one object or its own test targets, while a 16-kit carrier would remove about 23 t of rocket bodies. Filled markers: flown; open: planned or studied; sources in Table 7.

10.2 Trade space

To show how objects per mission scale with the two budgets, we re-ran the tuned beam search over a grid of ΔV budgets (1–5 km/s) and mission lives (3–15 yr) on 40 fixed starts. Figure 13 shows the medians. The count is limited by whichever budget binds: with 3 yr it stalls at 14–17 objects whatever the ΔV, and with 1 km/s it stalls at 14–15 whatever the time. At 7 yr the count rises from 15 (1 km/s) to 28 (2 km/s) and only to 33 (4–5 km/s), so above about 2 km/s the chemical propellant grows faster than the count. Long lives pay only with large budgets: 15 yr and 5 km/s give 63 objects. A visit-only servicer is therefore best sized at about 2 km/s for a 7 yr life, which is the small envelope of the draft.

10.3 Propulsion: chemical or electric for the drop-and-return profile

Each leg drops 25–700 km, waits, and returns, so the propulsion choice sets both the propellant and the time spent manoeuvring. A storable bipropellant gives I_{sp} = 300–340 s and a Hall thruster 1,500–2,500 s [43, Table 17-4, p. 692]; we keep the stated 320 s and 1,800 s. For the best small-envelope tour (36 objects, 1,995 m/s, 2,397 d), the electric servicer needs 108 kg of propellant

Table 7: Servicing and Removal Demonstrations Compared with the Drift-Phased Carrier

Mission (agency)	Year	Wet mass	Target	Capture	Objects	Cost, published
RemoveDEBRIS (EU) [4, 20]	2018, flown	n/f	own CubeSat targets	net, harpoon	2 own	EUR 15.3 M total (EU 7.0 M)
ELSA-d [4]	2021, flown	175 kg†	own client, ≤20 kg, prepared	magnetic plate	1 own	n.p.
ADRAS-J (JAXA) [34, 35]	2024, flown	180 kg	H-IIA upper stage, about 3 t	none (inspection)	0	n.p.
ADRAS-J2 (JAXA) [34]	FY2027, planned	n/f	same upper stage	robotic arm†	1	JPY 13.2 bn†
ClearSpace-1 (ESA) [21–23]	2029, planned	580 kg	PROBA-1, 95 kg, unprepared	capture and blocker arms	1	about EUR 100 M (ESA 86 M)
e.Deorbit (ESA study) [24]	to 2018	1,600 kg†	one large ESA derelict	arms, net or harpoon studied	1	n.p.
UK national ADR (UKSA) [69]	about 2029†	n/f	two UK-licensed defunct satellites	n/f	2	about GBP 61 M†
GEO life-extension servicer [4]	2020, operational	2,326 kg†	cooperative GEO client	docking	1 at a time	n.p.
This study: carrier with kits	concept	3.3–5.9 t	16 SL-8 bodies, 1,435 kg each (A)	one kit per object	16 (8–36)	566 M USD (2024, estimate)

† secondary source (database, trade press or filing; see `research/missions_v4.json`). n/f: not found in the sources reached; n.p.: not published. Commercial operators are not named.

against 622 kg, and its launch mass falls from 1,322 to 1,008 kg despite the heavier stated dry mass (900 against 700 kg). The price is thrust time. With one 1.35 kW, 83 mN thruster [43, Table 17-11, p. 707] firing in sunlight only, the transfers take 6,370 thruster hours, 17% of the mission, against a demonstrated life of more than 5,700 h for that class [43, p. 707]; the large envelope needs 17,450 h, or about three thruster lifetimes. The continuous-thrust tours of Table 6 reached 25.5–28 objects against 28, so the spirals cost about one object in this model. For the carrier the answer reverses. Its only large manoeuvre is the 209 m/s descent, the electric version saves 173 kg of propellant but is heavier at launch (3,309 against 3,282 kg), and the spiral down takes about 153 d, which delays every kit (Fig. 15). We therefore recommend chemical propulsion for the carrier and its kits, and electric propulsion only for a visit-only inspection tour, with at least two thrusters.

10.4 Capture, guidance and relative navigation

A rocket body offers no docking fixture, may tumble, and has a known geometry (nozzle, payload-adapter ring, cylindrical tank). The options demonstrated or planned are a net and a harpoon (RemoveDEBRIS), a magnetic plate on a prepared client (ELSA-d), robotic arms with blockers (ClearSpace-1) and an arm grasping the payload attach fitting (ADRAS-J2, secondary source) [4, 22]. For kits on tumbling bodies, the option that needs the least synchronisation is a contact attachment to the adapter ring or nozzle after the kit has matched the body's motion, or a harpoon; a net suits small objects but not a 1.4 t body that must then be pushed. We do not size the capture mechanism; this is the largest open item of the design. The navigation sequence follows the demonstrations: angles-only far-range tracking with a visible camera, mid-range lidar and thermal imaging, and close-range pose estimation from 3D lidar, the kind of vision-based navigation that RemoveDEBRIS tested [4]. These sensors, the kit interfaces and the capture

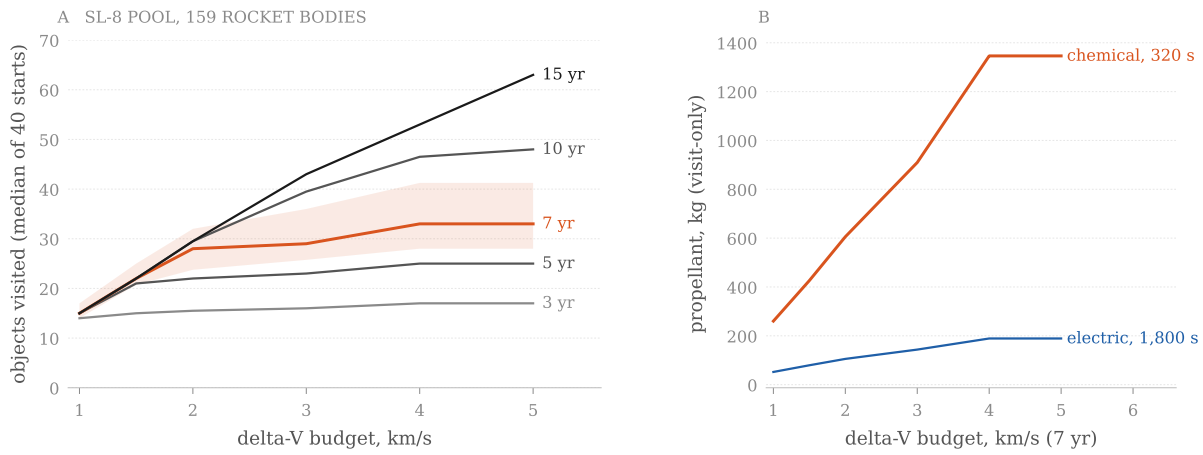


Figure 13: Objects visited against ΔV budget and mission life (median of 40 starts; band: interquartile range at 7 yr), and propellant for the 7 yr case. Both budgets must grow together. SL-8 pool, frozen snapshot, beam width 12, tuned weights.

hardware form the "payload" of the servicer in the budgets below.

10.5 Communications and operations

The drift waits are long and quiet, but the approaches are short and critical. At 975 km the longest pass over a station with a 10° mask is $T_{max} = P\lambda_{max}/180° = 12.4$ min [43, Eq. (5-52), p. 121], where P is the period and λ_{max} the maximum Earth central angle, and 8.2 min at the 573 km parking orbit. A numerical contact simulation for the 82.9° orbit gives 4.5 passes and 43 min per day for a station at 40° latitude, with gaps of up to 11.5 h, against 13.8 passes, 143 min and gaps of at most 1.6 h for a station at 78°. Even with a high-latitude station the servicer is in contact for less than 10% of the time, so the final approach, capture and kit release must be autonomous, with the ground approving each phase (go/no-go) rather than flying it. During the drift the operations are light: an element-set refresh and a small trim per leg, which Section 8 showed recovers four objects over open-loop execution. A two-station network, one at high latitude, and a small team are enough; we assume 12 staff-years per year (A) for the cost estimate.

10.6 Power and mass budgets

To size the power system we prepared an operating budget for three modes (A): drift cruise, proximity operations with kit release, and electric thrust. The loads, with a 25% margin for a new design [43, Table 10-9, p. 316], are 362 W, 775 W and 1,712 W. The array follows

$$P_{sa} = \frac{P_e T_e / X_e + P_d T_d / X_d}{T_d}, \qquad P_{EOL} = P_0 I_d \cos\theta \, L_d, \tag{2}$$

[43, Eqs. (11-5)–(11-8), pp. 413–417], where P_e and P_d are the loads in eclipse and daylight, T_e and T_d the eclipse (35.0 min at $\beta = 0$) and daylight durations, $X_e = 0.60$ and $X_d = 0.80$ the path efficiencies for peak-power tracking, $P_0 = 301\,\mathrm{W/m^2}$ for 22% multijunction cells, $I_d = 0.77$, $\theta = 23.5°$ and $L_d = (1 - 0.005)^{\text{life}}$ [43, Tables 11-34 to 11-36, pp. 412–414]. The battery capacity is $C_r = P_e T_e / (\mathrm{DOD}\, n)$ [43, Table 11-40, p. 422] with a depth of discharge of 25%, the upper end of the 15–25% that LEO cycling allows [43, p. 420]. For a chemical servicer the array is 7.9 $\mathrm{m^2}$ and 65 kg (25 W/kg, [43, Eq. (10-13), p. 333]) and the battery 2.0 kW h and 22 kg at 90 W h/kg;

for the electric servicer the array grows to 11.9 m^2 and 98 kg if the thruster fires in sunlight only. Table 8 gives the mass allocation, obtained by applying the average subsystem fractions of SMAD Appendix A [43, Table A-2, p. 895] to the stated dry masses, and Figure 14 the launch masses. The bottom-up array and battery (87–108 kg) are well below the 224 kg allocated to power, so the stated 700 kg dry mass leaves room for the capture hardware; the fractions come from communication satellites and are a check, not a design.

Table 8: Servicer Mass Allocation and Launch Mass

	chemical carrier	electric carrier
RPO suite, capture and kit interfaces (27.4%)	192	247
structure (21.3%)	150	192
thermal (3.6%)	25	32
power (31.9%)	224	288
TT&C (4.8%)	34	43
ADCS (6.9%)	48	62
propulsion (3.8%)	27	34
dry mass (A)	**700**	**900**
16 kits, dry 135 kg each (A)	2,160	2,160
kit rendezvous propellant	210	210
carrier propellant	212	39
launch mass, draft kits	**3,282**	**3,309**
launch mass, kits sized for 25-yr life, 5-yr life, re-entry	4,810; 5,099; 5,854	–

kg. Fractions: average of SMAD Table A-2 (communication satellites), normalised; masses from the rocket equation with the ΔV of Section 9.

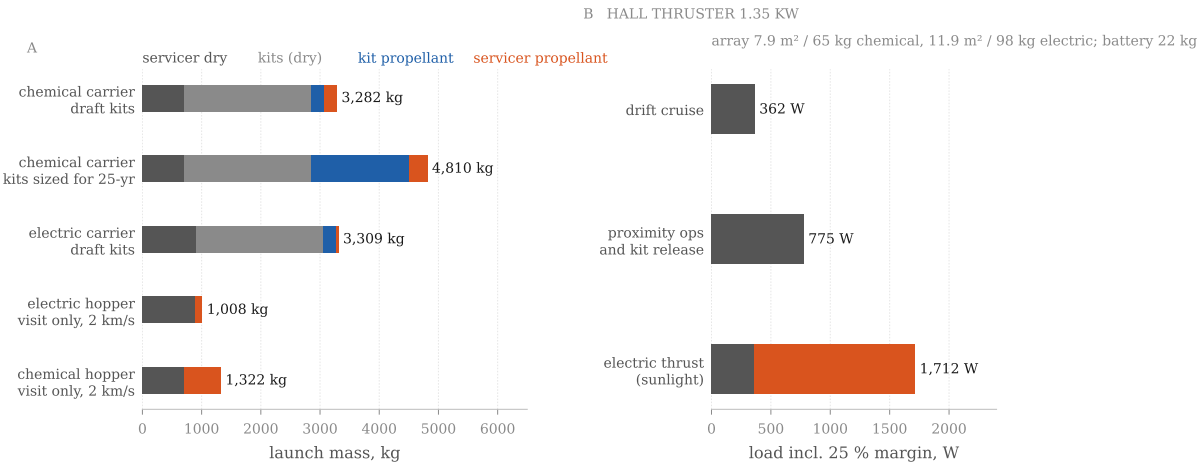


Figure 14: Launch mass by design and electrical load by mode. The kits, not the servicer, set the launch mass; electric thrust more than doubles the array power. Loads are assumptions (Appendix B); sizing after SMAD Sec. 11.4.

10.7 De-orbit kits

The draft fixed the kit at 135 kg and charged it only the rendezvous ΔV (about 254 m/s, 13 kg of propellant per kit). A kit must also lower the rocket body. We computed the perigee that gives a residual life of 25 or 5 years with an orbit-averaged drag model (same atmosphere as Section 2.1, density scaled by 0.5 and 2), using the median radar cross-section of the 159 SL-8 bodies (4.34 m^2) as a proxy for the mean projected area, $C_d = 2.2$ (A) and 1,435 kg (A). A single

burn from 975 km to perigee h_p costs $\Delta V = v_c - v_a(h_p)$. The 25-yr rule needs $h_p = 417$ km (371–466 km over the density bracket), 146 m/s and 81 kg of propellant at $I_{sp} = 280$ s; the 5-yr rule needs 314 km, 175 m/s and 97 kg; and a controlled re-entry with a 50 km perigee needs 251 m/s and 137 kg. The de-orbit propellant alone is therefore 60–100% of the draft's kit mass, and with it each kit weighs 237–298 kg at departure. For 16 kits the carrier's launch mass rises from 3.3 t to 4.8–5.9 t (Table 8), a factor of 1.47–1.78. The time advantage of the carrier is unchanged, but its mass advantage over a hopper (Table 6) shrinks accordingly, and the kit becomes a small spacecraft of its own with propulsion, avionics, navigation and a capture device.

10.8 Reliability over 7–15 years

For a series system $R = e^{-\lambda t}$, and n identical parallel strings give $R = 1 - (1 - R_1)^n$ [43, Eqs. (19-1), (19-3), (19-6), pp. 765–766]. No sourced failure rate for such a servicer was found, so we bracket $\lambda = 0.02$–0.2 per year (A). A single string then survives 7 yr with probability 0.87–0.25 and 15 yr with 0.74–0.05; dual strings raise this to 0.98–0.43 and 0.93–0.10. SMAD's data show that complex, programmable and actively controlled subsystems fail several times more often than simple ones [43, Table 19-4, p. 769], which is the case for an autonomous servicer. The architecture matters more than the redundancy. Since a failure ends all later placements, the expected number of kits placed is $\sum_k e^{-\lambda t_k}$. For the median start, the carrier places 15.9, 15.7, 15.5 and 15.0 of its 16 kits at $\lambda = 0.02$, 0.05, 0.1 and 0.2 per year, and the serial hopper 15.5, 14.7, 13.6 and 11.7. The 36-object visit tour of 6.6 years delivers 33.7 to 19.9 expected visits over the same range. Short campaigns are the most effective reliability measure for this mission.

10.9 Mission timeline

Figure 15 shows the timeline for the median start. After an assumed development of about five years (the value used by [66]) and 90 days of commissioning (A), the chemical carrier descends in about an hour and releases its kits as the drift brings each plane into line, the 16th at about 310 days after launch. An electric carrier releases the same kits about 150 days later, and a hopper that carries them needs about 1,320 days. After the last release the carrier must itself be disposed of: its parking orbit at 573 km has a natural life of about 17 yr in our model, which meets the 25-yr rule but not the 5-yr rule, for which a 36 m/s burn to a 442 km perigee is needed (149 m/s for a controlled re-entry).

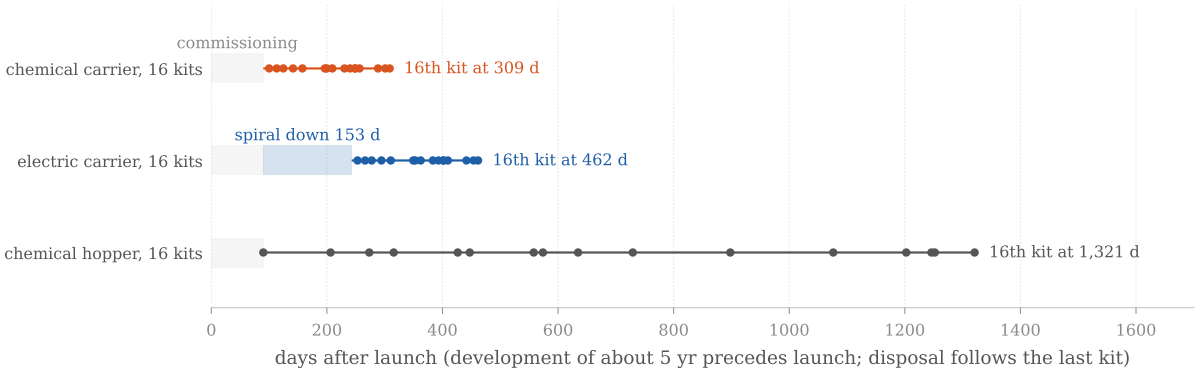


Figure 15: Kit release times after launch for the median start. The parked chemical carrier is done in about 310 days; the spiral of an electric carrier and the serial hopper add 150 and about 1,000 days. Commissioning (90 d) and development are assumptions.

10.10 Cost estimate

The cost was estimated with the parametric relations of SMAD Chapter 20, in fiscal-year 2000 dollars and converted to 2024 dollars with the consumer price index (factor 1.822 from annual averages of 172.2 and 313.7 [74]). The method is as follows. The bus (dry mass minus the 27.4% payload share) uses the large-satellite cost relations, 101 X for development and 43 X for the first unit, with X the dry mass in kg and standard errors of 33% and 36% [43, Tables 20-4, 20-5, pp. 795–796]; the RPO and capture payload is taken as 0.4 times the bus, the small-satellite convention [43, Table 20-6, p. 797], applied here beyond its range. Integration, program level, ground support equipment and launch support follow the same tables. Each kit is costed as a 135 kg small satellite ($781 + 26.1X^{1.261}$ k\$, standard error 3,696 k\$), split 60/40 into non-recurring and recurring cost [43, Table 20-9, p. 799], with production of N units costing the first unit times $L = N^B$, $B = 1 - \ln(100/S)/\ln 2$, and a 90% slope for 10–50 units [43, Eq. (20-5), p. 809]. Software costs 435 k\$ per thousand lines in flight and 220 on the ground (60 and 40 thousand lines, A), the ground segment is three times its software [43, Tables 20-10, 20-11, pp. 800–801], and operations cost 160 k\$ per staff-year plus 10% of the ground equipment per year [43, Table 20-12, p. 801]. Launch is priced at a published rideshare list price of 350,000 USD for 50 kg plus 7,000 USD/kg [56] (secondary, 2026 dollars, not inflated). The uncertainty of each element combines its standard error with a technology term from its readiness level, $\sigma = (\sigma_c^2 + \sigma_t^2)^{1/2}$ [43, Eq. (20-1), p. 805], with TRL 7 for the bus and 4 for the capture payload and kits ($\sigma_t = 10$ and 20%, [43, Table 20-15, p. 804]); the system σ is the average of the fully correlated sum and the uncorrelated root sum square [43, p. 805]. Table 9 gives the result.

Table 9: Parametric Cost Estimate, Chemical Carrier with 16 Kits (2024 M USD)

Element	cost	σ (%)
bus development / first unit	93.4 / 39.8	34 / 37
RPO and capture payload, development / first unit	37.4 / 15.9	39 / 41
integration, assembly and test	43.2	47
program level	62.4	40
ground support equipment	22.1	35
launch and orbital operations support	6.2	43
kit development	14.7	34
kit production (16 units, 90% learning)	102.9	34
flight software / ground segment	47.5 / 48.6	34 / 34
operations (1.3 yr)	9.0	32
launch (3,282 kg, rideshare list price)	23.0	–
total (system σ 133 M; 10–90%: 396–736 M)	**566**	23
serial hopper with 16 kits (3.6 yr of operations)	590	23
electric visit-only hopper, 2 km/s, 7 yr	559	25

Management reserve (about 20% [43, p. 803]) not included. Kits sized for re-entry add 11–18 M of launch cost.

The estimate is about 566 M USD (10–90%: 396–736 M), or 35 M per object and about 24,700 USD per kilogram of rocket body removed. Three points follow from Table 9. First, development dominates: the operations and launch that distinguish a carrier from a hopper differ by only 24 M, so the faster carrier is also no more expensive. Second, the 16 kits cost about a fifth of the total, and they are the element with the least heritage. Third, a visit-only tour costs about as much as a 16-kit removal mission and removes nothing. For comparison, the NASA cost-benefit study gives 3,000–40,000 USD per kilogram for a tug to uncontrolled re-entry [14] (dollar year not stated), ClearSpace-1 costs about EUR 100 M for one 95 kg object [21], and RemoveDEBRIS cost EUR 15.3 M for a demonstration on its own targets [20]. The estimate rests on cost relations derived from satellites of the 1990s, and the capture payload and kits have no direct analogue, so

it is a cost class (several hundred million dollars) rather than a price.

11 The Original Nanotug Concept: Full Engineering Analysis

Sections 9 and 10 moved away from the mission the 2025 study actually drew. This section returns to it. An ESPA-class electric-propulsion carrier holds 8U nanotugs, waits on a lower orbit until J_2 brings its plane onto the next derelict, raises to rendezvous and releases tugs. Each tug has a sticky side, attaches, detumbles the derelict, orients it and moves it. The 2025 deck assumed 40 tugs, two per object, 20 objects, 2,000 m/s and the cluster at about 975 km and 82°. We analysed that concept as drawn, in the order a systems engineer would, and we say plainly what fails. Every number below comes from a script in `analysis_v5/` (Appendix A) and every input carries a tag (M, D or A) in Table 16.

The main results are as follows. The tour and the storage close. The removal does not. Two tugs carry too little impulse to bring a 1.4 t stage down by a useful amount, and they were never designed to: the nano-tug paper the concept rests on [51] keeps the derelict in orbit and dodges collisions with it. The concept closes as a stabilise-and-dodge mission, and it closes as a removal mission only with a different tug.

11.1 Target selection by future collision risk

We first ask which objects a mission should visit. The tours of Section 7 maximise a count. To rank by consequence we use the catalogue rate model of Section 2 with two corrections: a 25 km shell and a collision area equal to the mean projected area of the real body rather than its radar cross-section. For each derelict we compute the rate λ of catastrophic collisions (threshold 40 J/g), the fragments $\geq$10 cm each one makes [37], and the residual orbital lifetime. Three metrics follow: expected fragments over a 25-year horizon, $R = \lambda N_{10} \min(25, L)$; the probability-times-mass form of the massive-derelict rankings [53]; and an index of the Anselmo–Pardini type [2] built from mass (exponent 1.75, our assumption), spatial density and lifetime.

Two facts about the pool come first. The "159-object SL-8 pool" of the earlier sections holds 139 SL-8 stages, 2 SL-14 stages and 18 payloads, so "159 SL-8 bodies" was loose wording. Inside it the risk spread is small: the 90th percentile is only 2.6 times the 10th, since most members share one mass, one area and one shell. The first two metrics are the same ranking by construction, and the third agrees with them at a rank correlation of 0.80. Median involvement rate is 1.2×10^{-4} per year per object; the median object carries $R = 4.8$ fragments and the whole pool 844.

Table 10 and Fig. 16 compare the objectives on the moving ephemeris of Section 11.2. With a 300 km drop-off, 7 yr and the budget of Section 11.3, count maximisation reaches 20 objects in 1,491 d and carries $R = 111.8$; risk maximisation reaches 19 objects, takes 2,413 d and carries $R = 132.5$, a gain of 19%. Thus within this band the risk objective trades time for risk at a modest rate; choosing well inside a nearly uniform pool cannot do more. The larger choice is the band.

The same vehicle and budget give $R = 117$ with 20 objects in the 83° band, 341 with 13 SL-16 stages at 71° and 840 km, 187 with 9 derelicts in the sun-synchronous pool, 28 with 11 objects in the Iridium-class pool and 176 with 9 objects in the mixed pool. The SL-16 band therefore offers about 2.9 times the SL-8 risk per mission, driven by 9 t stages. We keep the SL-8 band as the baseline for two reasons: the concept was drawn for it, and a 9 t stage multiplies the tug problem of Section 11.5 by six. The sun-synchronous and mixed values lean on the $m = 150A^{1.5}$ mass model and should be read as an upper range. Beyond that, the wider pools are justified only as a second mission; a single carrier cannot change inclination bands.

Table 10: Count and Risk Objectives on the Moving Ephemeris

Case	Objective	Objects	Risk R	Time [d]	ΔV [m/s]
5 yr, drop-off 300 km	count	20	111.8	1,491	1,326
5 yr, drop-off 300 km	risk	20	121.1	1,786	1,338
7 yr, drop-off 300 km	count	20	111.8	1,491	1,326
7 yr, drop-off 300 km	risk	19	132.5	2,413	1,376
7 yr, direct to 975 km	count	20	112.9	1,342	1,408
7 yr, direct to 975 km	risk	20	140.1	2,450	1,520

Beam search, 48 starts, cap of 20 objects from 40 tugs. R in fragments $\geq$10 cm avoided over 25 yr if the objects were removed.

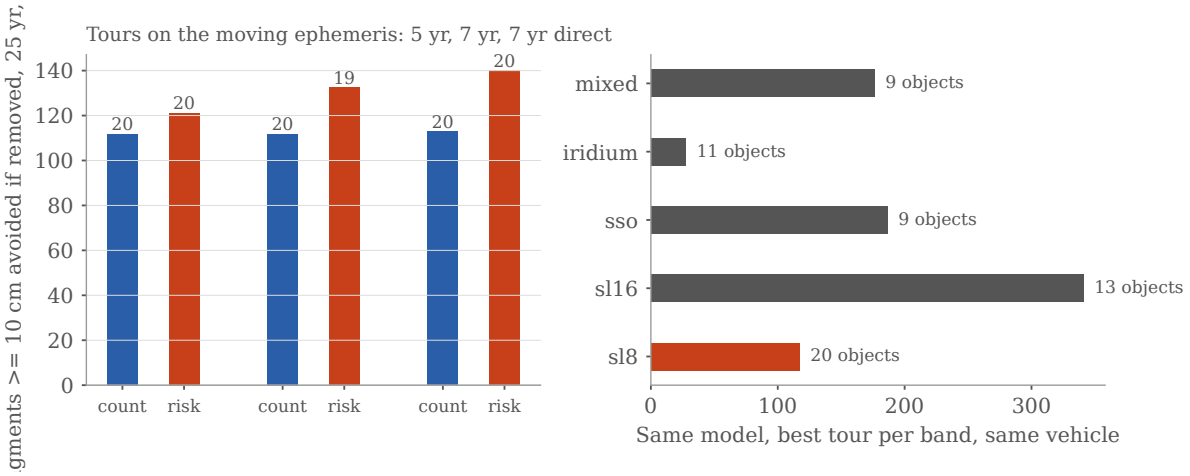


Figure 16: Risk removed by count and risk tours, and the best risk tour per band with the same vehicle. The risk objective adds about a fifth in the SL-8 band; the SL-16 band offers about three times as much per mission. Values from `q1_risk`.

11.2 Everything moves

We then replaced the frozen rate tables by a closed-form mean-element ephemeris. At 975 km and 82.94° the first-order nodal rate is 0.744°/day in magnitude and the second-order rate with J_2^2 and J_4 is 0.742°/day. The luni-solar node term is 4.4×10^{-5}°/day, four orders below it. Drag lowers an SL-8 body by about 123 m/yr at mean activity, 0.9 km over the mission, which adds a quadratic term to the node. The argument of perigee circulates every 128.6 d, and J_3 offsets the eccentricity vector by 1.0×10^{-3}, so the e-vector of a target is a moving goal; we charge its match in every spiral.

Against a full Cowell run (zonals to J_4, drag, radiation pressure, Sun and Moon) over 180 d on eight objects, the differences in nodal slope between objects have a standard deviation of 3.2×10^{-4}°/day, or 0.12°/yr, and inclination drifts by at most 0.008°. Differences are what matter, since the planner uses node gaps. The model is good for planning and needs element refresh for flight. The snapshot alone is not good enough: over the 5.4 yr from the snapshot to our assumed launch date (2032) the median pairwise node gap changes by 5.5° and the 90th percentile by 15°. Sequencing must therefore use where the objects will be, as the deck itself said it should.

The in-plane phase, omitted so far, turned out to be coupled to the node. Depth changes both the node rate and the mean motion, and both accumulate as the integral of the altitude offset. At this altitude one degree of node goes with 2,823 degrees of phase. Depth and dwell therefore cannot close both: the node closure fixes the phase. The phase laps every 3.5 d at 100 km depth and every 14 d at 25 km, so a shift of the dwell can close it, but this leaves a node error. We

remove that error with a small inclination offset held during the dwell, which works because the node rate changes by 0.105°/day per degree of inclination at this inclination, so the offset needed is only thousandths to hundredths of a degree. On realistic sample legs the numerical solution closes node and phase and costs a median of 0.5 m/s, at most 5.1 m/s. In-plane phasing is cheap, but not for the reason usually given.

11.3 The chaser

The bus is an ESPA-class electric bus with the published capability of its class [19]: 2,000 m/s, more than 1,920 kg of payload, more than 4 kW at end of life. Thrust comes from Hall thrusters of the 1.35 kW, 83 mN, 1,600 s class [43, 67], two firing and two spare, 166 mN for 2,700 W. Eclipses are at most 35 min of a 105 min orbit (fraction 0.33), and over the 208-day beta-angle cycle the orbit is in full sun 24% of the time. Firing in sunlight only with a 5% allowance for slews gives a mean duty of 0.76 (minimum 0.63). The SMAD array sizing [43] needs 4,557 W for two thrusters and the bus, 1.14 times the end-of-life allocation, so late in life the stack throttles.

Table 11 and Fig. 17 give each transfer both ways. The ΔV of a low-thrust spiral [17] equals the Hohmann cost to 0.03% at these depths, so the deck's budget stands. The time does not. A 100 km round trip is 101 m/s in both, but a "burn" at 166 mN lasts 8.5 d, or 116 orbits, and the round trip takes 22 d instead of 1.7 h. The spirals themselves drift the node by 0.41° at 100 km, which the planner credits.

Table 11: Hohmann and Low-Thrust Transfers from 975 km

Depth [km]	ΔV Hohmann [m/s]	ΔV spiral [m/s]	Spiral time [d]	One Hohmann burn [d]	Days per degree of node
25	25.1	25.1	5.5	2.1	112
50	50.3	50.3	11.1	4.2	56
100	101.2	101.2	22.3	8.5	27
200	204.4	204.4	45.1	17.1	13
300	309.9	309.9	68.4	25.9	9
500	527.6	527.7	116.5	44.1	5

Round trips, 2.4 t stack, 166 mN, duty 0.76.

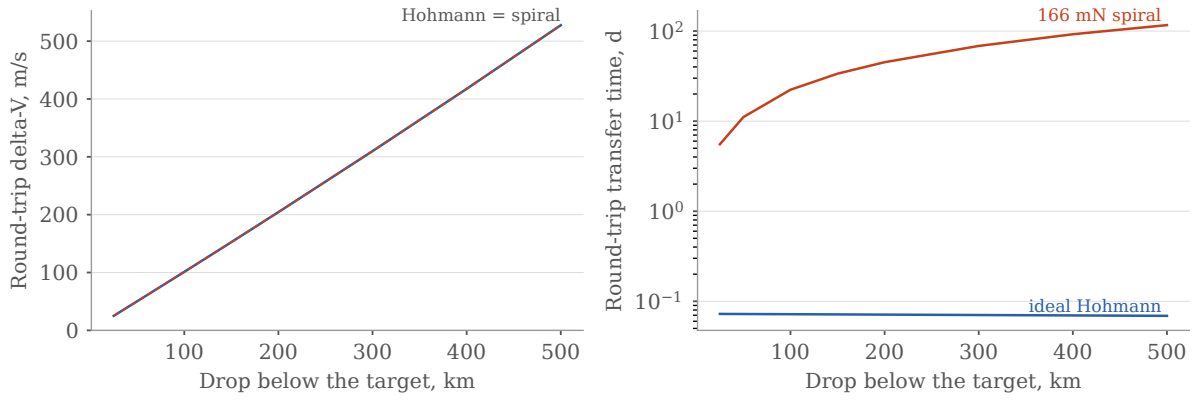


Figure 17: Each transfer as an ideal Hohmann and as a real 166 mN spiral. The ΔV is the same and the time is hundreds of times longer. Values from `q3_chaser`.

Launch and drop-off. The deck drops off at 300 km. A spiral from there costs 363 m/s and 80 d, the node slips 12° on the way (which the launch azimuth absorbs), and drag at 300 km is a fifth of the thrust (thrust to drag 4.8). From 600 km it costs 195 m/s and 43 d. No rideshare goes

to 83°, and getting there from a sun-synchronous rideshare would cost 1,928 m/s, so the plane has to be bought with a dedicated launch. Direct insertion at 975 km frees 363 m/s; at the same budget it makes the risk tour about 140.1 against 132.5, which suggests that a launcher with a restartable upper stage is worth more than extra propellant.

Budget. Of 2,000 m/s, insertion takes 363 m/s, disposal 198 m/s, and we hold 60 m/s back for replanning, leaving 1,439 m/s minus margin for the tour. Disposal uses the 5-yr rule: the empty carrier at area-to-mass of the array needs to spiral to 595 km (198 m/s, 13 kg of xenon, 18 d); the 25-yr rule would allow 703 km and 140 m/s.

Proximity operations. The carrier arrives on a relative ellipse of 5 km and holds on e/i-vector separation, which never crosses the velocity axis, so a loss of thrust leaves a passively safe drift. An ellipse of size ρ costs $n\rho$, 5.0 m/s at 5 km. Closing, entering, shrinking to 1 km and departing cost 17 m/s per object, close to the 15 m/s used earlier. The keep-out sphere is 200 m around a 6 m tumbling body, and only tugs go inside it. The carrier's abort speed from the keep-out boundary is 0.20 m/s.

Conjunctions. At the spatial density of the 950–1,000 km shell the carrier sees about 31 approaches within 1 km per year. With an assumed 2% needing action that is 0.6 actions per year and 0.01 m/s/yr, negligible in ΔV. It matters only in timing, since a Hall stack needs 1 to 3 days of warning.

Radiation. At 975 km the orbit sits on the flank of the inner proton belt. Secondary data points give 1 krad/yr at 800 km and 6 krad/yr at 1,400 km behind 5 mm of aluminium; log interpolation gives about 1.7 krad/yr at 975 km, and thin shielding (1–2 mm) multiplies it by 3 to 10, or 5–17 krad/yr. The carrier takes about 12 krad in 7 yr behind 5 mm, which is within reach of rated parts. An 8U tug built from commercial parts rated at 10–30 krad lasts between 0.6 and 6 yr. This is a bracket and not a design value: we did not find a published dose-depth curve at this orbit, and an AP-9/AE-9 run is the first thing a real study should do. It already shows that the 13-year tug life of Ref. [51] needs rated parts or more shielding.

11.4 Storage and dispenser

The 2025 capacity note claims 4 tugs per port, 5 ports, "two rows", 40 tugs. We checked it against the ring user's guide [54] and the canister data [65]. Its port data are right: a 15 in port has a 24 × 28 × 38 in box and 567 lb (257 kg) at a centre of gravity of 20 in. Three things are not. First, its "8U" box, 9.4 × 8.9 × 14.4 in, is the 12U payload envelope of the commercial canister: 19.8 L, whereas a true 2 × 2 × 2 unit is 11.6 L. Second, 12 kg is not the mass of this tug: Ref. [51] gives 17.7 kg and our bottom-up gives 23.6 kg (Section 11.5). Third, "two rows" does not map to hardware, since a commercial canister ejects one payload and a rear canister cannot fire through a front one. With commercial canisters the count is 20, not 40.

Forty does fit with a purpose-built tandem tube. Table 12 and Fig. 18 show the design. Four tubes, 250 mm square with rails, wing allowance and walls, cover a port face inside 25 mm of rattle space. Each tube is 623 mm long and holds two true-8U tugs, and three deep does not fit (910 mm against 890 mm available). With 23.6 kg tugs a port carries 239 kg, which passes the 257 kg rating on the mass-acceleration-curve basis and its moment (88 against 131 kg m), but exceeds the 220 kg heritage 8.5 g figure; with 17.7 kg tugs it is 192 kg and passes both. Five ports then give 40 tugs and 1,197 kg, well inside the bus's payload. The sixth port carries the sensor head. If stowed arrays or plume fields take one more port the count is 32.

Table 12: Tube Options per 15 in Port (Bottom-Up Tug Mass)

Option	Tubes	Tugs per port	Port mass [kg]	Tube length [mm]	Result
12U box, 1 deep	4	4	136	475	passes both
12U box, 2 deep	4	–	–	901	too long
12U box, 3 deep	4	–	–	1327	too long
true 8U, 1 deep	4	4	127	336	passes both
true 8U, 2 deep	4	8	239	623	passes MAC only
true 8U, 3 deep	4	–	–	910	too long

Port box 610 × 711 × 965 mm less 25 mm rattle space and 75 mm interface. Limits 220 kg heritage, 257 kg MAC.

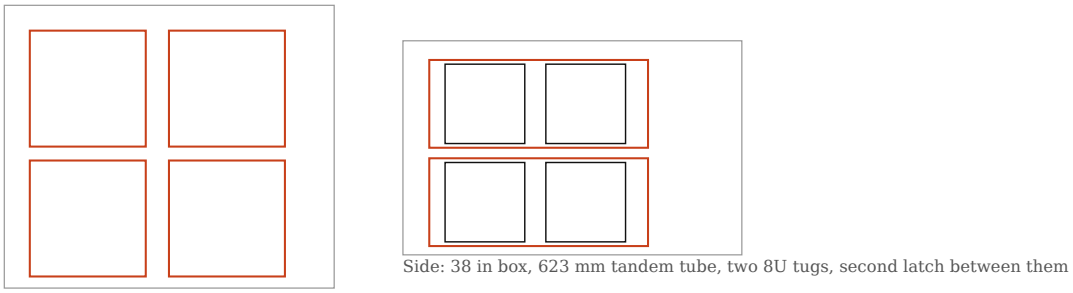


Figure 18: The published port box with the tandem tube design. Four tubes of two tugs each fit where the 2025 note drew four boxes. Values from `q4_dispenser`.

Restraint and release. Rail tabs are clamped by the door preload as in a canister. Each tug has its own restraint, because a single rail would see 3.9 kN at 8.5 g against the 3.6 kN tab rating of the canister: the rear tug rests on a mid-tube latch with its own non-explosive release, 1,967 N per latch. The front tug leaves first. The spring is slow on purpose, 4.2 N mean over 250 mm for 0.3 m/s, slower than a canister's release, so the tug stays near the carrier and spends little to null it. A commercial canister quotes under 10 deg/s of tip-off. With a CoM offset of 10 mm an unguided release gives 20 deg/s, and rails that hold the tug until the last 30 mm give 1.2 deg/s, which the tug's wheels take easily.

Centre of mass. Releasing tugs port by port moves the stack's centre of mass by 184 mm; a balanced order, front tugs first, holds it to 76 mm. At 166 mN a 76 mm offset would saturate wheels in hours, so the thrusters need 4.3° of gimbal. Access is through the door: rear tug, latch, front tug, door; a tube is removable as a unit.

11.5 Nanotug design

We re-derived the tug from Ref. [51]. Three points in that paper change how the deck should be read. The tug weighs 17.7 kg, not 12 kg, with 8.0 kg for propulsion including 4 kg of propellant, 5.3 kg of batteries and 4.4 kg for everything else. It exceeds the 16 kg an 8U CubeSat would be allowed at 2 kg/U. And its mission is not removal: the paper states that the key concept is to leave the derelict in orbit and perform just-in-time collision avoidance, with about 350 manoeuvres over 13 years. The 2025 deck turned this into removal.

Table 13 gives our bottom-up budget, using a listed 100 W, 7 mN, 1,000 s Hall thruster [7] with a mass of 1.16 kg. Four kilograms of propellant, three heads (one normal, two tangential), a cold-gas terminal module, a 300 W h battery, sensors and a pad come to 16.3 kg dry before

margin, 19.6 kg with a 20% margin and 23.6 kg wet. Volume is tighter than mass. Xenon at 1.6 kg/L takes the interior to 9.6 L against 7.2 L usable, and only 1.3 kg fits. Iodine, which the same thruster family accepts, brings it to 7.2 L, so a 4 kg tug closes in 8U only on iodine.

Table 13: Bottom-Up Nanotug Budget

Item	Mass [kg]	Volume [L]	Tag
structure, 8U frame	1.60	0.50	A
3 x Hall thruster, 7 mN class (1 normal, 2 tangential)	3.48	0.84	M
power processor, one shared (switched)	1.00	0.60	A
xenon feed, valves	0.50	0.30	A
xenon tank (25 % of load)	1.00	3.59	A
cold-gas terminal RPO module with 0.25 kg propellant	0.80	0.80	A
battery, 300 W h at 200 W h/kg	1.50	0.75	A
deployable arrays, 4 wings, 0.6 m2	1.20	0.50	A
power conditioning	0.40	0.25	A
3 reaction wheels, star tracker, IMU, Sun, magnetometer	1.50	0.80	A
OBC, S-band, UHF, GNSS, antennas	0.95	0.50	A
RPO camera and time-of-flight lidar	0.50	0.30	A
adhesive pad, load-sharing tendons, suspension, release	0.80	0.50	A
thermal, harness	0.70	0.25	A
spot shielding of electronics (2 mm Al equivalent)	0.40	0.15	A
Dry, with 20% margin	19.6		D
Wet with 4 kg propellant	23.6		D

Compare 17.7 kg in Ref. [51] and 12 kg in the 2025 note. Volumes are ours.

The conflict resolved. Section 10 found that a kit needs 81–137 kg of chemical propellant to lower a stage. A tug has 4 kg. Higher exhaust velocity closes part of that gap and no more. Two tugs at 1,000 s give the 1,435 kg stage 54 m/s (48 m/s with 160 kg of residuals, 74 m/s at 1,390 s). Spent as a spiral, that is 869 km and a residual life of 427 yr against about 1,073 yr today. Spent at apogee, it is a perigee of 765 km and 360 yr. Fig. 19 shows the result: a two-tug pair cuts the lifetime by a factor of two or three and leaves it in the hundreds of years. With the deck's 12 kg tug, carrying about 1 kg of propellant, the pair delivers 13 m/s.

A correction to Section 10 is needed here. It used the median radar cross-section, 4.34 m^2, as the drag area. A tumbling 6 × 2.4 m cylinder presents 13.6 m^2 on average. With the true area, the 25-yr perigee rises from 416 km to 500 km and the burn falls from 147 to 124 m/s. A 25-yr life then needs a spiral to 645 km, 171 m/s (146–195 m/s over a factor of two in density either way), and a 5-yr life 542 km, 227 m/s.

Removal by tugs alone therefore needs 7 tugs per stage for a 25-yr life (8 with residuals, 5 at 1,390 s) and 10 for 5 yr. Forty tugs then treat five or six stages at 1,000 s, not 20. Low thrust cannot give a controlled re-entry, and Ref. [51] notes that objects above 1 t will likely need one.

Power and thermal. Four wings of 0.6 m^2 give 197 W at peak. On a wall of a gravity-gradient stage, with fixed wings and hull shading, the orbit average is 38 W, and after 18 W of idle load 20 W is left. A 100 W thruster then runs at 20% duty, 1.4 mN on average, and the 4 kg lasts 1,557 h of firing (39 kN s) over 325 d. That is within the thruster's predicted life and it means power, not propellant, sets the pace: a pair takes about 453 d to deliver 40 m/s. A radiator face of 2U × 2U runs at 117 °C at full thrust in steady state and 70 °C at the duty above, so bursts are limited by thermal mass.

What a tug can do. A pair can do what Ref. [51] says. With two days of warning, a 1 km miss takes 1.9 mm/s and a 6 km miss 11.6 mm/s. The 6 km case takes 1.7 g of propellant, 20 min at

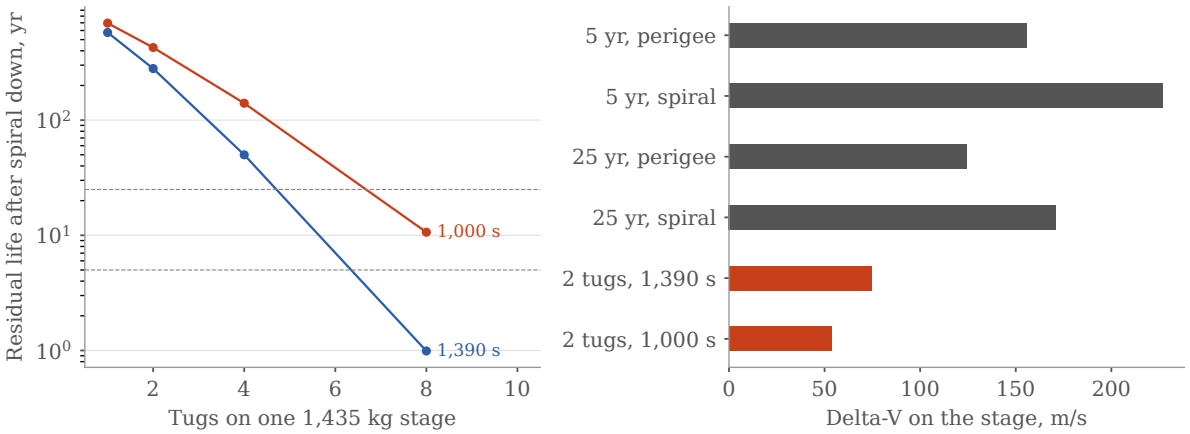


Figure 19: What tugs do to a 1,435 kg stage. A pair leaves hundreds of years of life; a 25-yr life needs 7 tugs at 1,000 s or 5 at 1,390 s. Values from `q5_nanotug`.

14 mN and 68 W h. The paper's 5-minute manoeuvre needs a mean 82 mN at 1,000 s, or 6 of these thrusters per tug, which neither the wings nor the 100 W class can support, but the same impulse over 20 min is fine. Its count of about 350 events implies 16 g per event including re-orientation. That count is propellant-limited; the radiation bracket and the storage life will likely limit it first.

The answer to the question of what goal is achievable is thus: stabilising and dodging with a pair, yes; a modest lifetime cut with what is left at end of service, yes; removal with a pair, no.

11.6 Attachment

We next take the sticky side at face value. The stage is an aluminium-alloy cylinder of 6 m by 2.4 m [68]. Its exact alloy and skin are not in any public source we opened. NASA's laboratory models were gray and white-and-orange [15], so the wall is painted. After decades at 975 km the binder will have aged. LDEF showed binder loss and surface texturing in atomic oxygen and ultraviolet [75], though atomic oxygen is orders of magnitude weaker at 975 km than at LDEF's altitude and ultraviolet is not. No measurement of a 40-year-old rocket body surface exists, and this is the largest unknown in the study.

The candidate is a gecko-type dry adhesive. Published patches give 5.1 kPa of adhesion with 17 kPa of shear on 1 cm^2 [60], and maxima about 20 and 80 kPa in the best cases [12]. They have run 30,000 cycles [60], full vacuum and −60 °C on over 30 spacecraft surfaces, with one year of continuous hold [61], and have grasped large free-floating objects in microgravity [36]. Scaled grippers perform well below patch values: the flight perching gripper measured 3.15 N [12]. Rocket-body capture with such adhesives has been studied [8], not demonstrated. Other options were judged weaker: pressure-sensitive adhesives are one-shot and outgas, electrostatic adhesion on a charged old surface is unproven, and clamps exceed 8U.

Hold is not the limit. A 0.03 m^2 pad at a 5% derate holds 7.7 N in tension and 25.5 N in shear. Centripetal load at 3°/s and 3 m from the axis is 0.19 N, and thrust is 0.03 N. Once seated, the tug stays seated. The tiles must, however, conform to a 1.2 m radius, a gap of 2.3 mm across 15 cm, which needs the load-sharing tendons of the grasping work.

Contact. Ground heritage is 1–2 cm/s for a 10 kg free-flyer [12]. Our contact model, a damped 30 mm suspension with a 3 mm preload stroke, gives a capture window of 1.4 to 13 cm/s at

200 N/m even at the 5% derate, and below the lower bound the tug does not preload the adhesive and rebounds. A window this wide is a model result; heritage still sits at its bottom end.

Temperature. A tumbling gray wall swings between -39 and 13 °C, and a white one down to -51 °C, inside the tested range. A wall that stays shaded falls to about -115 °C, outside it. Contact must therefore be made on a sunlit wall, which the attitude survey can arrange.

Where to touch. A tumbling stage is close to flat spin: Cowardin et al. found no spin-stabilised signature and Ojakangas and Cowardin expect a state near flat spin [15, 59]. The spin axis pierces the wall at two poles where the surface does not move, so a tug coming in along the angular momentum vector sees a turning surface and no translation. It cannot stay there, however: at the pole the lever arm is along $\mathbf{H}$ and no thrust can reduce $|\mathbf{H}|$. The pad must sit a distance ρ off the pole, and ρ trades authority against surface speed.

Rates. Measured synodic periods of unstable SL-8 passes run from 115 to 968 s [68], or 0.37 to 3.1°/s if each period is one rotation and half that if the light curve has two peaks per rotation, with a median of 0.56°/s. Three of ten passes read stable. The sample is small (seven periods, one site), and earlier notes quoting 3°/s as a median and over 10°/s as a tail for this class are not supported by anything we could open. Fig. 20 sets these rates against three limits at $\rho = 0.5$ m: tracking with two Hall thrusters, 2.0°/s; tracking with the cold-gas module, 5.3°/s; and seating within 5 s inside a 10 cm box, 2.3°/s. Attachment survives about 2.3°/s, which covers 86% of the measured rates, and moving to $\rho = 0.3$ m covers them all at the cost of slower detumble. Nutation moves the pole by about 13 cm, which is within pad size with closed-loop tracking.

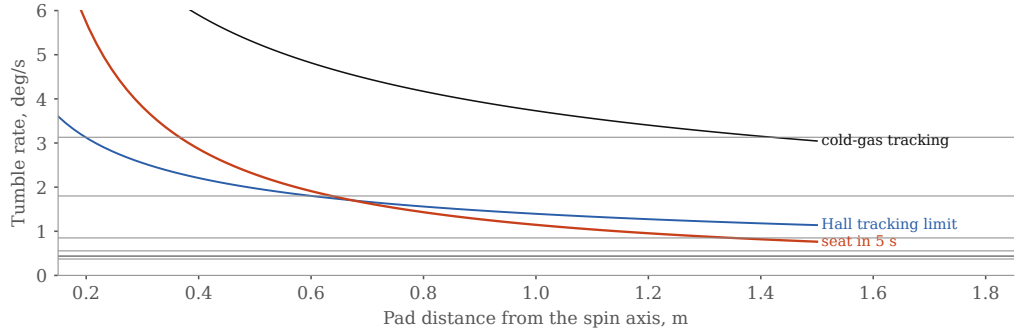


Figure 20: Tumble rate the attachment survives against pad distance from the spin axis, with measured SL-8 rates as hairlines. Near 0.5 m most measured rates are inside the seat limit. Values from `q6_attach`.

11.7 Detumble and orient

With $I_t = 5{,}300\,\mathrm{kg\,m^2}$ and $I_t/I_a = 4.3$ [59], the stage holds 93 N m s at 1°/s and 287 N m s at 3.1°/s. We simulated the rigid body with two tugs on the same wall at $\pm\rho$ along the axis. Opposed tangential thrust is a pure couple about the spin axis; together with normal thrust the pair has three-axis authority. Firing is limited by the 300 W h battery and the 20 W recharge. Fig. 21 shows the results. From 1°/s the pair stops the body in 3.6 h with 18 g in total; from the median rate in 2.0 h; from the fastest measured rate in 42 h with 58 g, with the battery as the limit. A nutating case with a 10° cone takes 17 h, 33 g and leaves a net 0.10 m/s. At $\rho = 0.15$ m the 1°/s case takes 45 h. The paper's case of 1.5°/s per axis costs 62 g in 45 h here against 17.4 g in 1 h there; the paper has longer arms and full power on more thrusters, and on our 100 W budget the time is set by energy. Either way detumble costs a few hundredths of a kilogram of the 8 kg. The

CAD model added one constraint: at 1 m spacing the 0.75 m wings of the pair overlap, so each tug seats rolled 20° about the wall normal, which costs 6% of tangential thrust.

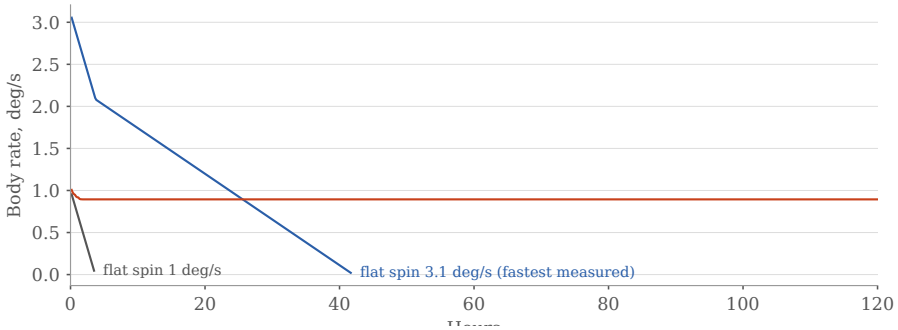


Figure 21: Detumble with two tugs, limited by the battery, and with one failed tug, which does not converge. Values from `q7_detumble`.

The failed-tug result is new and it matters. One tug on the wall cannot detumble the stage. Its tangential thrust at the wall also rolls the body about its long axis, and I_a is small; after 480 h and 254 g the rate is still 0.89°/s. A second tug must arrive, so spares matter. Thrust from one tug near the end of the body would be a pure transverse torque, but that spot moves fastest and it is out of reach for a Hall-only tug.

Thrust axis. The disposal or avoidance attitude is the gravity-gradient one: long axis along the local vertical, tugs on the $+V$ wall, on either side of the centre of mass, exhausting along $+V$. This has three advantages. The plume leaves the hull, the thrust presses the pad into the wall rather than peeling it, and gravity gradient supplies the stiffness (0.012 N m/rad, libration period 69 min). An eigen-slew of 90° takes 0.6 h and 3.1 g. With on-off differential control the attitude stays within 2.0° for a CoM error of 25 cm; with no control it swings by 11° for 10 cm. If one tug fails during a burn, the survivor's lever arm drives a static 17° and a swing of 29°, a cosine loss of 12%, and the burn can continue at half thrust. Rates below 0.09°/s are captured by gravity gradient, which agrees with the capture strategy of Ref. [51].

Contactless preconditioning. The earlier notes put an eddy-current coil on the carrier. For a thin shell the eddy tensor lies between $\frac{\pi}{2}$ and π times $\sigma e R^3 L$ (derived; see Ref. [63] for the method). This gives a natural decay time of 0.2–0.4 yr in the Earth's field for a continuous 3 mm shell, which is short against the age of these stages and shows that the real shell conducts less than this bound. A 2 m aluminium coil of 300 W and 50 kg ($7,043\ \mathrm{A\,m^2}$) at 5 m takes 1,015 d from 3 to 0.5°/s, a 2 kW and 300 kg coil 25 d, and a $10^5\ \mathrm{A\,m^2}$ superconducting coil 5 d at 5 m and 86 d at 8 m. The whole angular momentum reacts on the carrier, and the coil must sit inside the keep-out sphere. As the measured SL-8 rates are within the tugs' reach, the coil is not needed for this class; it remains a fallback for fast tumblers, at TRL 2–3.

11.8 Whole-mission closure

We now put the pieces together for the stabilise-and-dodge mission that the analysis supports, and we set it against two removal variants.

Mass, power, link, data. The launch mass is 2,382 kg: bus 900 kg, tubes 253 kg, 40 tugs 944 kg and 285 kg of xenon, with 723 kg of payload margin on the published figure. Fig. 22 shows it. The carrier's S-band downlink at 2,715 km slant range has 18 dB of margin at 1 Mb/s to a

3.7 m dish. The tug to carrier link has 27 dB at 5 km. After the carrier leaves, each tug talks to the ground directly, with 31 dB at 9.6 kb/s and 22 dB at 64 kb/s before polarisation and pointing losses on a wall. An object visit produces about 3.3 Gbit, or 0.4 d of downlink at 1 Mb/s from a 78° station (144 min per day, gaps up to 1.6 h). Approach, contact and abort are therefore autonomous, with ground review between gates.

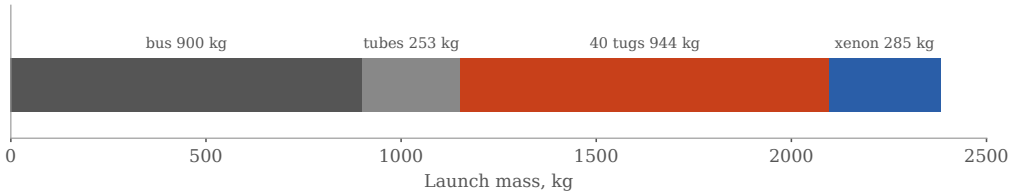


Figure 22: Carrier launch mass with 40 tugs at the bottom-up mass. Values from `q8_closure`.

Timeline. A stop takes 12 d: approach, a two-day attitude survey, release, transit, contact with a spare day, detumble, orient, handover. We flew the 7-yr risk tour leg by leg with numerical closure of node and phase. It visits 19 objects, ends with carrier disposal at 2,414 d (6.6 yr) and uses 1,928 m/s and 232 kg of xenon; the node closes to 1.0×10^{-9} deg and the phase to 0.00 deg. Thruster use is about 5,966 h per thruster against a rated 9,000 h. Figure 23 shows the mission.

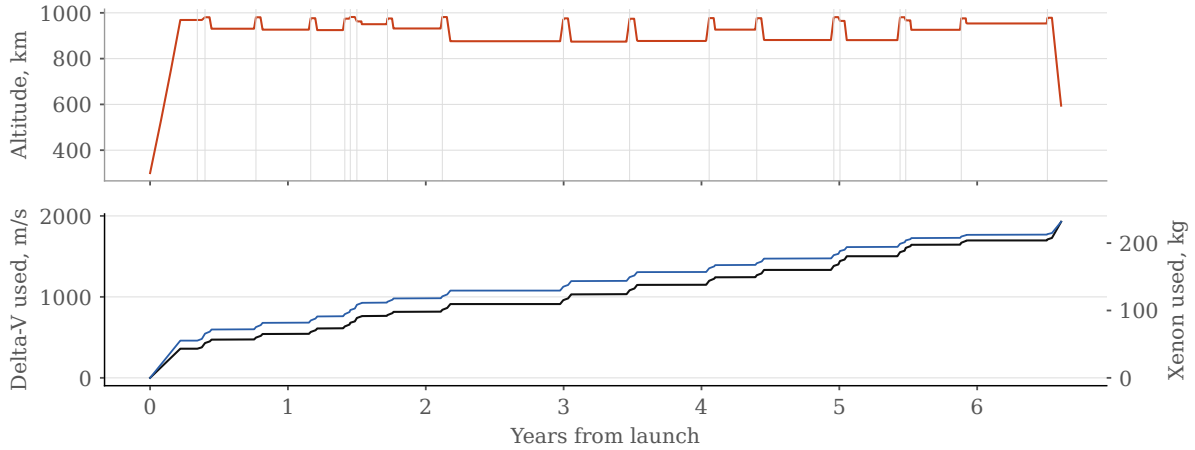


Figure 23: Carrier altitude and running totals of ΔV and xenon through the mission, from the numerical replay. Values from the replay file.

Success chain. A tug must be released, pass checkout after years in a tube, cross to the wall and seat; a pair must then detumble and orient. Only the rate data are measured; the gate values (release 0.99, checkout 0.97, transit 0.95, three seat attempts, detumble 0.92, orient 0.95) are engineering judgement. Fig. 24 gives the Monte Carlo with 4,000 draws per case and a fixed seed, with tumble rate drawn from the measured set, a rate limit of 2.3°/s and a bus failure rate. At 0.05 failures per year and 0.7 per seat attempt, two tugs per object and no spares give 8.8 stabilised objects of 19 visited. Allowing up to four tugs per object from the same 40 gives 11.9 (5–95%: 3–16), with a mean of 34 tugs used and 0.9 objects skipped for rate. The seat probability moves this from 11.0 to 12.3, the failure rate of 0.1 to 10.5, and a rate limit of 1 or 5°/s to 11.2 or 12.2. The 40 tugs therefore yield about a dozen objects and not 20.

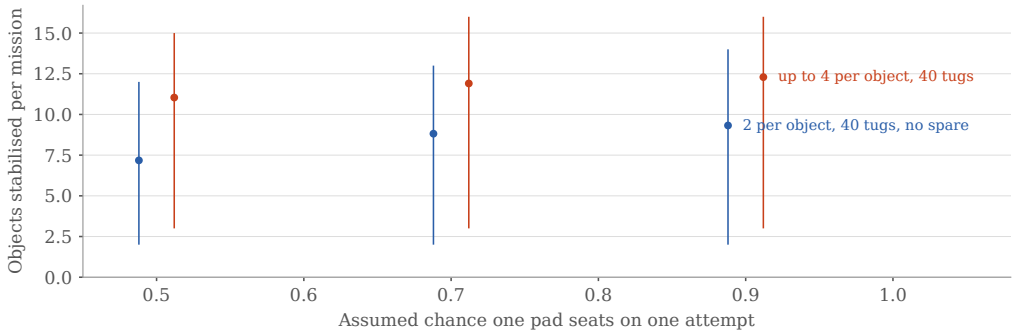


Figure 24: Objects stabilised per mission against the chance that a pad seats on one attempt, with 5–95% bars. Spares matter more than the seat probability in this range. Values from `q8_closure`.

Orbit dispersions. With thrust, I_{sp} and duty scattered and replanning corrections on each leg (assumed), the 95th percentile of tour ΔV is 1,441 m/s; the 60 m/s margin was set from this and 7% of draws exceed the unmargined budget. Series reliability over the tour is 0.88, 0.72, 0.52 and 0.27 at 0.02, 0.05, 0.1 and 0.2 failures per year, with 16.2 and 10.6 expected stops of 19 at 0.05 and 0.2.

Value. Table 14 sets the architectures side by side in the same risk units. Dodging is worth only what the tugs live through, times the fraction of events they catch: about 12 fragments avoided for a 7-yr life at 75% (range 3–28), or 0.007 catastrophic collisions avoided over 25 yr. The tug swarm treats 3.7 stages for 17 and 0.011. Paired electric kits, sized for the job at 41 kg each with a 600 W array and a 39 mN, 1,500 s thruster, fit 5 to a port by mass, treat up to 12 stages with 206 kg of xenon in about 63 d each, and give 34 and 0.021. Removal by kits is thus worth about three times the dodging mission, and it stays removed.

Table 14: What Closes and What Does Not

Architecture	Objects	Fragments avoided	Verdict	Why
40 tugs, 2 per object, removal (as drawn)	20 touched, 0 removed	0 from removal	does not close	pair gives 54–74 m/s; life stays 280–427 yr
40 tugs, 2–4 per object, stabilise and dodge	about 11.9	3–28	closes, if the pad seats	tour, storage, detumble, links close; service life and surface unproven
40 tugs, 7 per stage, 25-yr life	about 3.7	17	closes on paper	several attachments per stage, 324 d descent
Paired electric kits, 2 per stage	up to 7.0	34	closes on paper	not 8U; 600 W array per kit

Fragments ≥10 cm avoided over 25 yr, median-object rates, seat probability 0.7, 0.05 failures per year.

Cost class. With the SMAD relations of Section 10, a dedicated launch assumed at 75 M USD and 40 tugs at the learning slope for that quantity (41 M USD), the mission is about 837 M USD (1σ 192 M), or about 70 M USD per stabilised object and 49,000 USD/kg stabilised. This is a cost class. It uses a new-development bus relation, and buying an existing bus would lower it.

Against flown and planned missions. No demonstration in Table 7 has attached to an uncooperative rocket body, and none has placed more than one device on a third-party object. ADRAS-J inspected a stage of a similar class, so attitude survey of a tumbling stage is the

only step here with flight heritage. A single-stage, two-tug demonstration would buy down the adhesive risk before any 40-tug carrier is built.

Legal points. A derelict remains the object of its launching state. Under Article VIII of the Outer Space Treaty [72] the state of registry keeps jurisdiction and control, and 311 of the 317 intact objects in this shell carry one owner code, so touching one needs that state's consent, which any mission must obtain first. Article VI makes the tug's state responsible for authorising and supervising it, Article VII and the Liability Convention [46] attach liability, Article IX asks for due regard and consultation, and the Registration Convention [64] applies to each tug. A tug on a stage makes a joint object whose movements have to be coordinated with conjunction services. A US operator would also need licences for communications, for remote sensing of other objects, and for launch payload review. These are points for counsel; we offer no legal opinion.

12 Regulatory and Policy Context

The rules that govern disposal apply to the servicer and its kits, not retroactively to the targets. The IADC guidelines (Rev. 4, 2025) ask that LEO objects be removed by re-entry or moved to an orbit with a residual life "as short as practicable and no more than" 25 years, with a disposal success probability of at least 90% and a goal of 99% [29]; the UN COPUOS mitigation guidelines (2007) and long-term sustainability guidelines (2019) carry the same intent [70, 71], and ISO 24113:2023 is the top-level standard [30]. Two stricter rules now apply in practice. The US FCC requires disposal no later than five years after the end of the mission for satellites it authorises that are launched after 29 September 2024 [25, 26], and ESA's requirements of 2023 set an orbital lifetime below five years, a disposal probability above 0.9 and a collision-avoidance threshold of 10^{-4} per conjunction [18]. For this concept, the kits sized above meet either rule for the targets, and the carrier needs a 36 m/s burn to meet the five-year rule. A controlled re-entry would remove the casualty risk on the ground, for which the IADC limit is 10^{-4} per re-entry [29]; we did not compute the casualty area of an SL-8 body.

Two further constraints are legal and operational. The state of registry retains jurisdiction and control over its objects [72], and 158 of the 159 SL-8 bodies carry one owner code in the catalogue, so a removal campaign in this band needs that state's consent or participation. Damage in space is a matter of fault [73], which makes the close approaches of the servicer a liability exposure for its operator. The drift orbits also sit in the lower LEO shells, down to about 275 km, for months at a time, so the servicer must take part in routine conjunction assessment: screening against the public catalogue and conjunction messages, a manoeuvre threshold of the order of 10^{-4} as used by NASA's conjunction assessment practice [55] and ESA [18], and coordination through the civil traffic-coordination service that the US Office of Space Commerce plans to run in parallel with military services [58]. The planned close approaches to each target are themselves conjunctions and should be announced in the same way.

13 Discussion

13.1 Implications for multi-object removal

The physical core is simple. The nodal rate is a steep function of altitude, so a drop of 25–150 km buys rate differences of 0.01–0.06°/day for 25–150 m/s round trip, and the node of a prograde orbit at 83° then closes a few-degree gap in 30–300 d. The numbers add five points to the draft. First, the size of the advantage over impulsive plane changes is a function of the time allowed, and it is large only if years of bus life are available. Second, the benefit of choosing the sign per

leg is real (about two objects, and a factor of two in expected wait), so the unidirectional rule is not required. Third, the sequencing heuristic was not the weak point; the scalarisation weight and the best-of-N reporting were. Fourth, the visit count is not a removal count. With 135 kg kits, eight kits imply about 2.1 t at launch for a hopper, so removals per launch are mass-limited well below 29 or 67 objects, which is why the carrier-and-kits architecture is the one idea that matters. Fifth, in this model the collision-risk benefit of a 28-object removal from SL-8 is a few hundredths of a catastrophic collision over 25 years. The case for removal therefore rests on the heavy tail and on cascade dynamics, not on the first-generation average, and the model cannot settle it.

The servicer design of Section 10 adds three points. First, the time that drift phasing costs is also a reliability cost, and it is the strongest argument for the parked carrier: placing 16 kits in about 310 days keeps the expected loss below one kit even at a failure rate of 0.2 per year, while a serial hopper loses about four. Second, the mass and cost of a removal mission are set by the kits. Sized to lower a 1.4 t body, each kit needs 81–137 kg of de-orbit propellant on top of the draft's 135 kg, and the 16 kits are about a fifth of the cost. Third, at about 35 M USD per object the concept is below the cost per object of ClearSpace-1 (about EUR 100 M for one object), but it is not yet credible as a plan, because the capture of a tumbling, unprepared rocket body by a small autonomous kit has not been demonstrated by any mission.

13.2 Novelty assessment

Already known. Using J_2 drift for multi-debris tours, with drift orbits below, between or above the debris, is in Cerf (2013), with a lower-orbit worked example [9, 10]. The GTOC9 winners exploited it in branch-and-bound and beam searches [32, 62]. Raised waiting orbits with deployable kits are in [66], and a carrier with 2–8 kits waiting in a drift orbit 100 km below is in [13]. That altitude-lowering is more efficient than raising for nodal phasing is in [52], and the principle is patented prior art from 1993 [40]. Drift-orbit optimisation with drag in servicing is in [3], and with low thrust in [38]. Time-dependent TSP, evolutionary multi-chaser and dynamic-programming sequencing are in [27, 33, 50, 76]; risk-based ADR planning exists [28, 57]; and a recent preprint treats the drift orbit as a design variable and finds that inclination tilt beats altitude for near-sun-synchronous targets [31].

Not found in print (elementary, apparently unpublished). The closed-form ratio $R(x)$ with its remainder bound, the $1 + \frac{3}{2}x$ Hohmann ratio and $1 + 3x$ net gain, the decision threshold $180°(1 + \frac{9}{4}x)$ and the eccentric-target reversal. These follow from the convexity of $a^{-7/2}$ and small-parameter series, so their weight is small.

New evidence, not a new method. The audit and frozen snapshot; the weight correction and exact-optimum validation; the Cowell/SGP4 closure test; the quantified baselines; the risk benefit with its uncertainty; the novel-idea screen (ten ideas); the exact reproduction of the 2025 counts; and a subsystem-level servicer design with a cost estimate.

Candidate novelty. The persistent-drift-orbit carrier whose kits depart without a carrier return, quantified here but untested against a full kit design. We do *not* claim new astrodynamics, any removal capability, or any prediction of environment evolution.

13.3 Sources of error and model limitations

Several simplifying assumptions in this analysis warrant discussion regarding their effect on the results.

Leg model: The leg model is impulsive and circular-to-circular and has no in-plane phasing. Proximity operations and capture are one fixed allowance (15 m/s), which is unverified and decisive: an allowance of 50 m/s costs −8 objects.

Free insertion: Launch inserts the chaser at the start object for free, whereas a real launch inserts at its own plane and node.

Element-set errors and verification sample: Element-set errors beyond the assumed values are untreated, and the verification covers eight legs of one tour.

Atmosphere: The atmosphere is a tabulated exponential model (unverified, uncertain by a factor of 2–3). Solar activity is bracketed, not simulated.

Mass and kit assumptions: The mass and kit assumptions (700 kg, 135 kg, I_{sp}) are stated, not sourced.

Risk model: The risk model is a kinetic-gas estimate with 66% imputed cross-sections, no cascade feedback and no operational-asset weighting.

Pools and epoch: The pools are four bands of a 2026-08-18 catalogue. Results for other bands (especially near 600 km, where drag matters) are not claimed. No re-fetch of the catalogue was made, so the numbers describe that snapshot only.

Relaxation bound: The relaxation bound on the full pool is loose, so the full-pool optimality gap is not certified.

Literature coverage: The literature coverage is a bounded search, and items marked unverified in `research/sources.md` were not used.

Servicer budgets: The loads, the dry masses and the mass fractions are assumptions or communication-satellite averages, not a design. They place the servicer in the right class (hundreds of watts, about a tonne dry) and show that the kits dominate, but a heavier capture payload would raise the dry mass and, through the cost relations, the development cost roughly in proportion.

Kit sizing and lifetime model: The residual-life perigees use an exponential atmosphere, a radar cross-section as a proxy for area and a single burn; the density bracket moves the 25-yr perigee by ±45 km and the propellant by ±7 kg, so the conclusion that the draft's kit is too light holds. Kit avionics, capture hardware and tanks are still not counted.

Reliability: No sourced failure rate exists for this class of servicer, so the reliability results are a bracket over $\lambda = 0.02$–0.2 per year. The ranking of carrier over hopper holds over the whole bracket; the absolute values do not.

Cost: The cost relations were derived from 1990s satellites and are applied to a payload and to kits without analogues; the small-satellite payload factor is used beyond its range, the software sizes and staffing are assumptions, and the launch price is a secondary list price. The 10–90% interval reflects the published standard errors and readiness levels, not these model choices, so the true uncertainty is larger. The result is a cost class, and it should be compared with the demonstrations only at that level.

Version 5 model limits. **Surface:** no aged rocket-body surface has been measured, so the seat probability is a judgement that moves the result from 11.0 to 12.3 objects. **Radiation:** the dose is a bracket from secondary points; a real run could move the tug life either way. **Tumble data:** seven periods from one site; a wider survey could add a fast tail that needs preconditioning. **Bus data:** dry mass and propellant are earlier study values, not reopened here. **Mean elements:** the tour is a planning solution on a 2026 snapshot propagated to 2032; flight planning needs

fresh elements. **Chain gates and risk value:** the gates are judgement and the value uses median-object rates; neither changes the ranking of architectures, which rests on the ΔV result. **Thermal and power on the wall:** wing shading and thermal capacity are estimates; a tug with better sun geometry would be faster, not stronger.

13.4 What a reviewer would challenge

The strongest objections to this analysis, and our answers, are as follows.

1. *"Your gain over the draft is a weight change."* Yes: the draft's configuration was untuned. The comparison against the *published-style* fixed orbits (Table 3) is the relevant one for a literature claim.
2. *"The impulsive baseline is a straw man."* It is the best single-burn hop with an exact Hohmann and inclination charge, but it excludes multi-burn bielliptic options (tested and worse for small angles) and a plane change at higher altitude. Its time (16–36 d) is its real advantage, and a time-weighted objective could reverse the ranking.
3. *"15 m/s proximity operations and no phasing."* Correct, and decisive (Section 8). The count is conditional on that allowance.
4. *"Free insertion at the start object."* Yes; the starting-plane acquisition is not charged.
5. *"Counts are visits, not removals."* Yes; see Section 9.
6. *"Single pool, single epoch."* The pool is nearly exhausted at 5 km/s, and the SSO, Iridium-class and SL-16 bands appear in Table 4 only as medians.
7. *"Optimality gap not certified on the full pool."* Correct (Section 6).
8. *"Atmosphere, mass, kit and cost values are assumptions."* Yes. They are tagged in `research/params.json` (VERIFIED/DERIVED/UNVERIFIED); the headline numbers are insensitive to the atmosphere for this pool and sensitive to the proximity allowance.
9. *"The risk model is not validated."* It has no calibration to observed conjunction or fragmentation rates beyond the sanity check of 55 years between events if every derelict pair were involved, so it should be used for ranking only.
10. *"The cost estimate is a parametric guess."* Yes. It is a SMAD-style estimate with stated inputs and uncertainty, adequate to place the concept in a cost class and to show that development, not operations, dominates (Table 9).
11. *"Was the literature search complete?"* No. Several papers named in the brief (Han 2024, Yang 2024 as stated, Ruggiero, Lüthi) could not be verified and were not used.

14 Conclusion

This work re-evaluated nodal-precession phasing for multi-object debris removal on one frozen catalogue snapshot, reproduced the draft study's 2025 pipeline from its original inputs, checked the planner against an exact optimum and the leg model against high-fidelity propagation, and turned the tour into a servicer design with a cost estimate and its regulatory context.

At the level of the phasing geometry, lowering gains over raising by $R(x) = 1 + \frac{9}{2}x + \dots$ in rate and by $1 + 3x$ per unit ΔV. The gain is small (4% at 100 km) and one-signed, so the bidirectional rule is better. At the level of the planner, a tuned beam search is within 2% of optimal on small instances. At the level of the data, the draft's 63,553, 1,288 and 593 are reproduced exactly; they

come from two different 2025 catalogues and include objects that had already re-entered, and the comparable 2026 count of objects in orbit is 1,151.

At the mission level, a single spacecraft can visit a median of 29 SL-8 rocket bodies in 7 yr with 2 km/s, and 67 in 15 yr with 5 km/s, against 9 and 19 by impulsive plane change. Visits are not removals. The architecture that turns drift phasing into removal is a chemical carrier parked below the cluster that releases one kit per object; it places 16 kits in about 310 days, which also makes it the most reliable option, but kits that can actually lower a 1.4 t body make it a 4.8–5.9 t launch and a mission in the several-hundred-million-dollar class, about 35 M USD per object. Every flown or planned demonstration so far handles one object, and none has captured a tumbling, unprepared rocket body.

In general, the results demonstrate that drift phasing trades mission time for propellant rather than providing a time-free saving, and that its practical value lies in the number of objects per launch, which is limited by the kits rather than by the orbital mechanics. The main recommendations are as follows:

1. Plan removal around a parked carrier with kits and short campaigns, not a serial tour; the carrier is faster, more reliable and no more expensive.
2. Use chemical propulsion for the carrier and the kits; keep electric propulsion for visit-only inspection tours, with at least two thrusters.
3. Size the kit first: its de-orbit propellant (81–137 kg for an SL-8 body), its capture device and its autonomy set the launch mass and a fifth of the cost.
4. Fly a single-kit demonstration on one SL-8 body, combining the non-cooperative rendezvous already shown in orbit with a kit attachment and a perigee-lowering burn, before any multi-object campaign.
5. Report counts on a dated, hash-pinned catalogue with an explicit decay filter, and report tours as distributions over starts rather than as the best case.
6. Treat consent of the state of registry, conjunction coordination and the five-year disposal rule as design inputs from the start.

Future work should model the in-plane phasing and proximity-operations costs explicitly, design the kit and its capture mechanism, compute the re-entry casualty risk of an SL-8 body to decide between controlled and uncontrolled disposal, and extend the analysis to other bands and to a refreshed catalogue.

Acknowledgments

The author acknowledges the public satellite catalogue and element-set services that made the frozen snapshot possible, and the authors of the open-source propagation and scientific Python libraries used throughout this analysis.

The original concept was then analysed as drawn. A real ESPA-class carrier flies a tour of 19 objects inside 2,000 m/s including insertion and disposal, with low-thrust spirals that cost the same ΔV as Hohmann transfers and hundreds of times the time, and with in-plane phasing that is cheap once a small inclination offset decouples it from the node. Forty tugs fit only in purpose-built tandem tubes. The tug weighs 23.6 kg, not 12 kg, and a pair delivers 54–74 m/s to a stage, so two tugs per object is a stabilise-and-dodge mission, not removal. At mission level this closes for about 11.9 objects per carrier with spares, and removal closes only with 7 tugs per stage or with paired electric kits. In general, the results demonstrate that the 2025 orbital mechanics stand and that the 2025 end effector does not. Future work should measure an aged

rocket-body surface, run an AP-9/AE-9 dose at 975 km, widen the SL-8 tumble survey and fly one stage with two tugs.

A Reproducibility Statement

Snapshot. v3-2026-08-18: Celestrak SATCAT csv (70,292 rows) and GP three-line sets (SL-8 159, SSO 135, Iridium-class 28, SL-16 20, mixed 393), public, no credentials, fetched 18 August 2026; the SHA-256 of every file is in `data/SNAPSHOT.json`.

Code. `run_all.py` runs the tests, audit, analysis, figures and this PDF; version hash `800672fbc9995a1e` over 121 files (`CODE_MANIFEST.json`); Python 3.14, numpy 2.4, scipy 1.17, matplotlib 3.10, sgp4 2.25, sympy 1.14; no GPU, about 80 min on one core. This version was prepared on 4 November 2026.

Seeds. Start subsets 7, 9, 42; Monte Carlo 1, 2026, 11, 100+replicate; NSGA-II 0–2; bootstrap 1–2; every stochastic step is seeded inside its script.

Provenance. The original repository was read-only; its code is copied to `legacy/` and used only to reproduce its numbers. No data were invented, and unverified inputs are listed in `research/params.json`.

Version 4 additions. `analysis/a9_orig2025.py` reproduces the 2025 counts from `data/orig2025/` (Section 3.1); `analysis/a8_servicer.py` computes the servicer budgets, trade grid, reliability, kit sizing, timeline and cost (Section 10); `report/make_figures_v4.py` draws Figs. 12–15 for the paper and the site versions in `figs_site/` with their `manifest.json`. The mission data and their status (verified or secondary) are in `research/missions_v4.json`.

Version 5 additions. `analysis_v5/` holds `q1_risk` to `q8_closure`, the tour engine `tour5.py`, the replay `make_replay.py`, the figures and `tests/`. Inputs are in `params.py` with tags; the values in Section 11 are filled from `results/*.json` by `make_paper_v5.py`. Seeds are fixed in each script. `analysis_v5/run_v5.py` runs the whole chain.

B Servicer Parameters

C Inputs of the Nanotug Analysis

Table 16: Inputs of the Nanotug Analysis (M: Measured or Published, D: Derived, A: Assumed)

Name	Value	Tag	Source
sl8_mass	1435.0 kg	M	v3 value; Marchionne et al. 2021 give about 1,440 kg dry
sl8_mass_residual	1600.0 kg	M	Marchionne et al. 2021: 160 kg assumed residual propellant, total 1,600 kg
sl8_L	6.0 m	M	Stromberg et al. 2023 Table 1
sl8_D	2.4 m	M	Stromberg et al. 2023 Table 1
sl8_It	5300.0 kg m2	D	thin shell m(R^2/2+L^2/12) = 5,338 kg m2, rounded; earlier notes give 4,820-6,150
sl8_ratio	4.3	M	Ojakangas et al. 2012: I/I3 about 4.3 for SL-8

Name	Value	Tag	Source
sl8_inertia_unc	0.1	M	Marchionne et al. 2021: inertia, mass, CoM uncertainty about 10 %
sl8_skin_mm	3.0 mm	A	no public drawing found; aluminium-alloy skin thickness
sl8_sigma	17000000.0 S/m	A	Al-Mg alloy conductivity; alloy not confirmed by a public source
sl8_rot_periods	200, 968, 840, 808, 115, 424, 648 s	M	Stromberg et al. 2023 Table 2: synodic periods of the unstable SL-8 passes (table layout shifted; the set is what is used)
sl8_stable_of	3, 10	M	Stromberg et al. 2023 Table 2: 3 of 10 SL-8 passes read stable
flat_spin_residual_deg	6.0 deg	M	Ojakangas et al. 2012: with 1 % of maximum energy above flat spin the long axis is still 6 deg off
cd	2.2	A	standard drag coefficient for a compact body
mass_parus_class	825.0 kg	A	payloads (Cosmos, Nadezhda, Tsikada) in the 83 deg shell; no opened source
launch_iso	2032-01-01T00:00:00 UTC	A	about five years of development after this study; tours must be re-solved on fresh elements
dropoff_km	300.0 km	M	the 2025 deck: launch to 300 km
espa_box_in	24.0, 28.0, 38.0 in	M	Moog ESPA User's Guide, Nov 2018, Sec. 3e: 24 x 28 x 38 in box, 15 in port
espa_port_limit_kg	257.0 kg	M	Moog guide: 15 in port, 1/4 in fasteners, 567 lb at CG 20 in, MAC basis
espa_port_limit_heritage_kg	220.0 kg	M	Moog guide: 485 lb with heritage 8.5 g/8.5 g load factors
espa_cg_in	20.0 in	M	Moog guide: capacities quoted at CG 20 in from the port
espa_ring_od_mm	1575.0 mm	M	62 in ring (1,575 mm separation system, public bus sheet)
espa_ring_h_in	24.0 in	M	Moog guide: 15 in port ring is 24 in tall
bus_dv_ms	2000.0 m/s	M	public product-line sheet of an ESPA-class electric bus: 2,000 m/s
bus_payload_kg	1920.0 kg	M	same sheet: > 1,920 kg payload
bus_power_eol_W	4000.0 W	M	same sheet: > 4 kW EOL allocation
bus_power_bol_W	6800.0 W	M	same sheet: 6.8 kW BOL array
bus_life_yr	5.0 yr	M	same sheet: > 5 yr; 7 yr used in v3
bus_dry_kg	900.0 kg	A	v3 study value; datasheet not reachable this session
bus_xe_cap_kg	558.0 kg	A	earlier notes quoting the datasheet; not reopened this session
bus_area_m2	22.0 m2	A	two arrays about 10 m2 plus body; used for carrier drag
hall_F	0.083 N	M	SPT-100 class: 83 mN (SMAD Table 17-11; vendor listing)
hall_P	1350.0 W	M	SPT-100 class: 1,350 W
hall_Isp	1600.0 s	M	SPT-100 class: 1,600 s
hall_life_h	9000.0 h	M	vendor listing: 9,000 h guaranteed; SMAD >5,700 h demonstrated
hall_n_active	2	A	two firing, two cold spares: 2.7 kW of the 4 kW EOL
tug_hall_F	0.007 N	M	Busek BHT-100 listing: 7 mN
tug_hall_P	100.0 W	M	Busek BHT-100 listing: 100 W
tug_hall_Isp	1000.0 s	M	Busek BHT-100 listing: 1,000 s
tug_hall_mass	1.16 kg	M	Busek BHT-100 listing: 1.16 kg thruster
tug_hall2_F	0.013 N	M	Busek BHT-200 listing: 13 mN
tug_hall2_P	200.0 W	M	Busek BHT-200 listing: 200 W
tug_hall2_Isp	1390.0 s	M	Busek BHT-200 listing: 1,390 s
tug_ppu_mass	1.0 kg	A	power processing unit per firing thruster
mt_mass	17.7 kg	M	Marchionne et al. 2021 Sec. 3.4: 17.7 kg total
mt_prop_frac	0.45	M	same: 45 % of mass to propulsion, propellant included

Name	Value	Tag	Source
mt_batt_frac	0.3	M	same: 30 % to batteries
mt_prop	4.0 kg	M	same Sec. 8: 4 kg propellant per nano-tug
mt_n_per	2	M	same: two nano-tugs per derelict
mt_detumble_g	17.4 g	M	same Sec. 5.2: 17.4 g for 1.5 deg/s per axis, 1 h
mt_jca_g	2.5 g	M	same Sec. 6: 2.5 g, 5 min, 2 days ahead, about 6 km
mt_jca_n	350	M	same Sec. 8: about 350 maneuvers
mt_rpo_g	4.0 g	M	same Sec. 4: about 4 g for short-range RPO at 1 deg/s per axis
deck_tug_kg	12.0 kg	M	the 2025 capacity note: 8U CubeSat about 12 kg
cds_kg_per_U	2.0 kg/U	A	CDS-type allowance (12U = 24 kg, dispenser vendor paper); 8U = 16 kg
csd12_mass	5.65 kg	M	dispenser data sheet 2002337G: 12.46 lb empty
csd12_box_mm	239.0, 226.0, 366.0 mm	M	dispenser payload spec 2002367F (9.41 x about 8.9 x 14.4 in): the 2025 note's '8U' box is the 12U box
csd_tipoff	10.0 deg/s	M	dispenser data sheet: $<$ 10 deg/s per axis
csd_spring_N	11.6 N	M	dispenser data sheet: 2.6 lbf per ejection spring
csd_temp_op	-34.0, 80.0 C	M	dispenser data sheet, operational qualification
csd_wall_mm	12.0 mm	A	canister wall and rail allowance per side (external drawing not extracted)
u_mm	113.0 mm	M	CDS unit width class (226 mm for 2U)
rattle_mm	25.0 mm	A	dynamic rattle space per side inside the port box
wedge_normal_kPa	5.1 kPa	M	Parness et al. 2009: 5.1 kPa adhesion with 17.0 kPa shear, 1 cm2
wedge_shear_kPa	17.0 kPa	M	Parness et al. 2009
wedge_lab_max_normal_kPa	20.0 kPa	M	Chen et al. 2022: typical maxima about 20 kPa normal
wedge_lab_max_shear_kPa	80.0 kPa	M	Chen et al. 2022: about 80 kPa shear
wedge_cold_C	-60.0 C	M	Parness et al. 2013 abstract: tested in thermal vacuum to -60 C
wedge_cycles	30000	M	Parness et al. 2009/2013: 30,000 cycles
astrobee_force_N	3.15 N	M	Chen et al. 2022: 3.15 N measured
astrobee_contact_ms	0.02 m/s	M	Chen et al. 2022: 1-2 cm/s contact speed, 0.02 N/mm suspension
astrobee_k_Nmm	0.02 N/mm	M	Chen et al. 2022 suspension stiffness
aged_factor	0.05, 0.2, 0.5	A	derate on aged paint (no measurement on a 40 yr old surface exists)
e_star	40000.0 J/kg	M	catastrophic threshold 40 J/g, as v3
lc	0.1 m	M	10 cm fragments, NASA standard breakup model
horizon_yr	25.0 yr	A	as v3
f_warn	0.5, 0.75, 0.9	A	fraction of catastrophic-capable conjunctions warned and dodged by a live tug pair
rpo_hold_km	5.0 km	M	Marchionne et al. 2021: relative parking orbit of a few km
keepout_m	200.0 m	A	chaser keep-out sphere around a tumbling 6 m body
failure_rates	0.02, 0.05, 0.1, 0.2 1/yr	A	series failure-rate bracket, as v3

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Table 15: Inputs of the Servicer Design (M: Measured or Published, D: Derived, A: Assumed)

Parameter	Value	Tag	Source
dry mass, chemical / electric servicer	700 / 900 kg	A	draft design, `research/params.json`
specific impulse, chemical / electric / kit	320 / 1,800 / 280 s	A	within SMAD Table 17-4, p. 692
Hall thruster power, thrust, demonstrated life	1.35 kW, 83 mN, >5,700 h	M	SMAD Table 17-11, p. 707
kit dry mass	135 kg	A	draft concept note
SL-8 body mass	1,435 kg	A	`research/params.json`
SL-8 area proxy (median radar cross-section, 159 bodies)	4.34 m^2	M	2026 catalogue snapshot
drag coefficient	2.2	A	–
loads (cruise / proximity operations), before margin	290 / 620 W	A	guided by SMAD Table 10-9, p. 316
load margin	25%	M	SMAD Table 10-9, p. 316
cell efficiency, degradation	22%, 0.5%/yr	M	SMAD Table 11-34, p. 412
inherent degradation, Sun angle	0.77, 23.5°	M	SMAD Tables 11-34, 11-36, pp. 412–414
path efficiencies X_e, X_d	0.60, 0.80	M	SMAD p. 413
battery DOD, specific energy, efficiency	25%, 90 W h/kg, 0.90	A/M	SMAD pp. 420–422 (15–25%, 70–110 W h/kg)
array specific power	25 W/kg	M	SMAD Eq. (10-13), p. 333
subsystem mass fractions	Table 8	M	SMAD Table A-2, p. 895
failure rate bracket	0.02–0.2 /yr	A	no sourced value
commissioning; development	90 d; about 5 yr	A	[66] for development
flight / ground software	60 / 40 thousand lines	A	–
operations staff	12 staff-years per year	A	–
cost relations, learning slope, TRL sigma	as in text	M	SMAD Tables 20-4 to 20-15, pp. 795–809
TRL: bus / capture payload and kits / software	7 / 4 / 5	A	our reading of SMAD Table 20-15
CPI-U 2000, 2024 annual average	172.2, 313.7	M	[74]
rideshare list price	350,000 USD + 7,000 USD/kg	M	secondary [56]
ground station mask	10°	A	–

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